



Horizon Europe Programme

Standard Application Form

Project proposal: Stop Transmission of gambiense Human African Trypanosomiasis (STROGHAT)

Technical description

Abstract

Gambiense human African trypanosomiasis (gHAT) is a neglected tropical disease caused by trypanosome parasites. gHAT is fatal if left untreated. So far, treatment options for gHAT were limited and toxic, forcing control programs to avoid overtreatment through complex diagnostic procedures, including screening with a serological test, laborious microscopic confirmation of seropositives and lumbar puncture for disease stage determination. This resulted in loss of up to 50% of gHAT cases, which remained untreated.

Recently a non-toxic single dose oral drug, acoziborole, has shown 98.1% efficacy in a phase III trial, irrespective of gHAT disease stage. Acoziborole removes the need for lumbar puncture and appears safe enough to treat serological suspects without microscopic confirmation (*Screen & treat*). The STROGHAT project 1° will evaluate effectiveness of a *Screen & treat* approach to rapidly reduce gHAT prevalence in an entire focus; 2° will extend acoziborole safety documentation; 3° and will analyze costs of this new approach. To achieve these objectives, *Screen & treat* will be implemented, actively and passively, for 3 consecutive years in the gHAT focus of Nord Equateur in D.R. Congo. Available geographical information will be exploited to specifically target villages where gHAT was recently, or still is present. Detection at a reference laboratory, of the trypanosomes nucleic acids in blood collected before treatment, will retrospectively identify true gHAT cases among the treated serological suspects. After 3 years of intervention, the gHAT prevalence in the focus will be re-estimated. STROGHAT intends to provide the first evidence for recommending *Screen & treat* to national HAT control programs for elimination of gHAT. Through facilitated diagnosis, increased acceptability and access to treatment, STROGHAT will contribute to achieving the goal of stopping gHAT transmission by 2030, as defined by the World Health Organization.

Participant No.	Participant organisation name	Country
1 (Coordinator)	Prins Leopold Instituut voor Tropische Geneeskunde (ITM)	Belgium
2	Programme National de Lutte contre la Trypanosomiase Humaine Africaine (PNLTHA)	Democratic Republic of the Congo (DRC)
3	Institut National de Recherche Biomédicale (INRB)	Democratic Republic of the Congo (DRC)
4	Institut de Recherche pour le Développement (IRD)	France
5 (Associated Partner)	Drugs for Neglected Diseases Initiative (DNDi)	Switzerland

1. Excellence

1.1 Objectives and ambition

The **STROGHAT project** will uptake new treatment and diagnostics into clinical practice and evaluate effectiveness of implementing a *Screen & treat* strategy to rapidly reduce gambiense human African trypanosomiasis (gHAT) prevalence. It will extend safety documentation of acoziborole, a new single dose oral drug that will be used for treatment, and costs of this new strategy will be analyzed. To achieve these objectives, *Screen & treat* will be implemented, actively and passively, for 3 consecutive years in the isolated gHAT focus of Nord Equateur in the Democratic Republic of the Congo (DRC). Existing geographical information will be exploited to specifically target villages where gHAT was recently, or still is present. Laboratory detection of the trypanosomes nucleic acids in blood collected before treatment, will retrospectively identify true gHAT cases among the treated seropositives. After 3 years of intervention, the gHAT prevalence in the focus will be re-estimated. STROGHAT will provide the first evidence of this new approach, which will impact the World Health Organization (WHO) guidelines for gHAT control and consequently impact the national strategies of all affected African countries. STROGHAT will thereby contribute to achieving the goal of stopping gHAT transmission by 2030, as defined by WHO.

Background

Human African trypanosomiasis (HAT) or sleeping sickness is a vector borne parasitic disease, fatal if left untreated, affecting rural communities in sub-Saharan Africa.[1] The parasite is transmitted by flies of the genus *Glossina*. [2] There are two forms of HAT of which by far the most common is the form caused by the parasite *Trypanosoma brucei gambiense*, referred to as gHAT [3] The disease has two stages, of which the first stage has few and atypical symptoms. In the second stage, the parasite has crossed the blood-brain barrier and more typical, neurological symptoms arise, eventually leading to coma and death.

Since the 19th century, HAT is known to have caused several devastating epidemics. It is estimated that at the turn of the 20th century 300,000 to 500,000 people died of HAT, with another epidemic occurring in the 1920s-‘30s.[4] By 1960 intensive efforts had brought the disease under control again

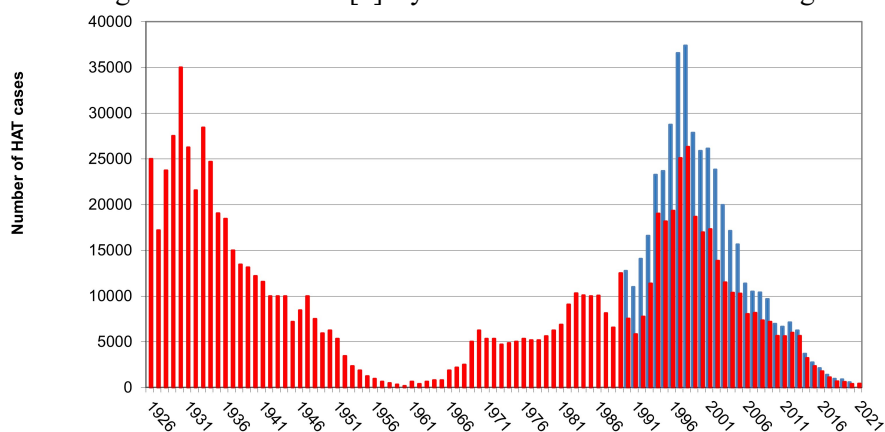


Fig. 1: gHAT case notification trends in DRC (red) since 1926 and continent-wide (blue) since 1990.

in most countries but when measures were neglected it reemerged. Hundreds of thousands have perished from HAT as recently as the 1990s.[5] Continent-wide trends are clearly driven by the DRC caseload (Fig.1).

In its first road map for the prevention and control of neglected diseases of 2012, WHO targeted HAT for elimination as a public health problem by 2020.[6] Following major innovations in diagnostics, treatment and vector control, the number of reported gHAT cases declined drastically and this goal can be considered as reached. [7] The backbone to reach these goals was based on active screening of

the population at risk via mobile teams, followed by treatment. In the DRC more than two million are routinely annually screened, with a peak of 3 million in 2019 whereby only 613 cases were detected.

Currently incidence is at an all-time low but this does not mean that interruption of transmission can be obtained by continuing the current approach. At present, most mobile teams only detect a few cases per year, if any. The alternative is passive screening based on detection of gHAT in patients attending fixed health facilities. However, health facilities in gHAT affected areas are typically poorly equipped and poorly attended, implying that passive screening alone cannot guarantee interruption of transmission. As fewer cases are found, diagnostic skills gradually disappear. Attendance rate is a major challenge for screening effectiveness but also other determinants play a role. Robays et al. estimated that up to 50% of prevalent cases are not detected or not cured (optimistic scenario), with major losses occurring in the parasitological confirmation step.[8] Fear of the lumbar puncture and of toxicity of treatment, in particular associated with melarsoprol, which the older population still remembers, are other important barriers.[9] Because humans are the reservoir for gHAT, not treating substantial numbers of infected persons will lead to sustained transmission and reemergence, as observed in the past.

Until recently treatment options for gHAT were limited and toxic, forcing control programs to avoid overtreatment.[10] An estimated five to ten percent of patients treated for second stage HAT died from the treatment, which was based on melarsoprol, an arsenic derivative. [11] This explains the complex diagnostic procedures including screening with a serological test followed by microscopic confirmation of seropositives and determination of disease stage, which required a lumbar puncture, on the spot. gHAT treatment is only administered after visualization of the trypanosome through a microscope, but this parasitological confirmation is complex and lacks sensitivity. The sensitivity of the mAECT, the most sensitive test available and thus the current confirmation test of choice is <90%. Since 2010, new and safer drugs have been introduced, i.e. intravenous nifurtimox-eflornithine combination therapy (NECT). More recently, fexinidazole, an oral 10-day treatment effective against the 2 stages of disease is available. [12] This was undeniable a big step forward, but some issues remain such as a 10 day compliance, obligatory administration under supervision, and inferior effectiveness in severe gHAT disease.

Recently a non-toxic single dose oral drug, acoziborole, has been developed by Drugs for Neglected Diseases initiative (DNDi) in collaboration with Sanofi, which has passed phase 3 evaluation.[13] It acts on the factor CPSF3 of the trypanosome, thereby specifically inhibiting trypanosomal mRNA processing, and this mode of action will prevent the occurrence of field resistance. [14] Acoziborole can treat gHAT irrespective of disease stage, thus rendering the lumbar puncture for stage determination redundant. Having a single dose oral drug available for both stages of the disease, opens up new perspectives for eliminating gHAT. Treating anyone testing positive to a serological screening test, without further diagnostic confirmation and stage determination (“Screen and treat” strategy), will avoid missing many true HAT cases and will reduce the complex procedures needed to confirm the infection. Between 1985 and 2004, a strategy for elimination of gHAT transmission based on Screen & treat, without confirmation, indeed has been proven successful in an island setting. [15] However, with treatment options available at that time, it was not feasible for large-scale application in mainland foci. With the arrival of acoziborole, a safe, single dose oral drug, that does not have to be administered in a hospital setting, Screen and treat could become widely applicable, provided the safety profile is confirmed with additional data to enable detection of uncommon adverse events.

Several clinical studies have been conducted over the last years to characterize safety, tolerability, elimination, metabolism and the pharmacokinetics profile of acoziborole (explained in detail in the clinical study information annex). A phase II/III, multicenter, open-label prospective study (DNDI-OXA02-HAT) has been conducted to assess the efficacy and safety of a single dose of 960 mg of acoziborole under fasting conditions in 208 patients with g-HAT, over 15 years of age, in DRC and Guinea. A total of 27 serious adverse events (SAEs) were reported in 21 patients exposed to acoziborole, among which 4 fatal cases. All SAEs were assessed as not related to acoziborole by the investigators and the sponsor. The most frequently reported SAEs were infections and infestations or psychiatric or central nervous system disorders. The results demonstrated a high efficacy 95.2%

(159/167; CI 95% Jeffreys: 91.2- 97.7) for the modified intention to treat set and 98.1% efficacy (159/162; CI 95% Jeffreys: 95.1- 99.5) in the evaluable set with a very good safety profile: only 14% of treatment emergent adverse events were related to acoziborole. Complementary analyses comparing the success rate of acoziborole obtained in this study with that of NECT, the current drug of choice for severe gHAT disease, in three previous studies, indicated that the success rates of acoziborole and NECT were similar regardless of disease severity.

An ongoing pediatric study (DNDI-HAT-OXA05), supported by EDCTP2, assesses safety and tolerability of a single dose in patients from 1 to 14 years in DRC and Guinea, whereby 35 children with gHAT will be treated with doses adapted to 3 body weight bands (10- 20; 20-30 and 30-40 kg).

An ongoing phase II/III study (DNDi-OXA-04-HAT) assesses safety and tolerability of a single dose of acoziborole compared with placebo during a follow-up of 4 months in gHAT seropositive individuals who are parasitologically negative (900 active; 300 placebo) in DRC and Guinea. The draft blinded safety result of 500 participants already treated are encouraging: 8 % of the participants presented adverse events (1.8 adverse events per subject) occurring within 3 days after administration and with a duration of 24 hours. Full results of this study will be available before the start of STROGHAT.

STROGHAT wants to assess the effectiveness of implementing the *Screen & treat* strategy, including its costing. Within this epidemiological study, a clinical study will be nested to contribute to the provision of more safety data on acoziborole.

Objectives

Primary objective:

- To evaluate if a *Screen & treat* strategy can lead to interruption of gHAT transmission in a mainland focus

Secondary objectives:

- To provide further evidence for safety of acoziborole in seropositive gHAT suspects
- To provide an accurate cost estimate for a *Screen & treat* strategy

Ambitions beyond the state-of-the-art

- Currently, effectiveness of *Screen & treat* has hardly been documented; **STROGHAT will** determine effectiveness in a gHAT, after a 3-year intervention (WP2&3&4).
- Available safety data of acoziborole are not sufficient to justify wider use without confirmation of disease; **STROGHAT will** for the first time generate safety data of acoziborole which is administered “on the spot” and thereby contribute to fill this prerequisite (WP1&3).
- Currently no evidence exists on cost of *Screen & treat*; **STROGHAT will** determine its cost, allowing control programs and funders to be informed and opt for the most effective HAT control strategy (WP3&6).
- Participation to screening is a major drawback, in particular to reach groups more at risk (e.g. adults males); **STROGHAT will** try to better understand potential barriers and adopt strategies to facilitate participation (WP5); **STROGHAT deals with** some barriers already identified (e.g. direct and opportunity costs linked with visit to health center and hospitalization, no need anymore to fear lumbar puncture, ...)
- The actual diagnostic chain results in treatment of 50-75% of gHAT patients only; **STROGHAT will** increase effectiveness by avoiding microscopic confirmation and lumbar puncture, i.e. previously additional obligatory steps before administration of treatment (WP3);

- The most sensitive parasitological confirmation techniques were accessible only to specialized and well-equipped health structures;
STROGHAT will increase access to treatment, by simplifying the diagnostic procedure, which can be performed by non-specialized staff (WP1&3).
- The National Reference Laboratory for HAT in DRC has remained so far under-used;
STROGHAT will prepare the National Reference Laboratory for HAT for its future central role in high throughput confirmation diagnosis of treated seropositives (WP2)
- So far laboratory standard operating procedures were accessible only for specialized staff;
STROGHAT will improve access to correct procedures for all health workers by making comprehensive video protocols of the most relevant procedures and rendering those public on the website (WP2&7)
- The rapid decline of gHAT prevalence has an impact on proficiency of lab staff in the field; Quality assurance of diagnostic tests in the field was limited to teams using the specialized “TRYPELIM tool” allowing digital recording of pictures and videos and; Actual prevalence data rely on highly specialized, laborious microscopic confirmation by mAECT, which has limited sensitivity;
STROGHAT will eliminate highly complex confirmation microscopy and centralize confirmation capacity at the national reference laboratory, meaning that less specialized staff is needed; thereby **STROGHAT will extend quality control for all teams and centers in the field** and; thereby **STROGHAT will** also generate more reliable data of prevalence than with the current lower sensitivity tests (WP2).
- The actual gHAT treatment algorithm is still complicated and requires hospitalization or at least daily consultation by health staff, even with advent of the oral drug fexinidazole;
STROGHAT will administer treatment on the spot, increasing coverage of gHAT treatment (WP1&3).
- The treatment strategy is currently based on under-treatment, justified by previous toxicity of available treatments;
STROGHAT will switch to over-treatment, which should allow to also treat cryptic infections in which trypanosomes are not present in the bloodstream (WP1&3).
- Currently cure of disease is difficult to confirm and requires a long post-treatment follow-up;
STROGHAT will for the first time implement RNA molecular tests in the field, demonstrating clearance of infection [17] (WP2&3).
- The current gHAT control strategy is not fit for purpose in insecure areas;
STROGHAT provides opportunities to quickly move in and out of insecure areas and ensure treatment (WP3).
- Lack of reliable infection data is a leading cause of uncertainty in models assessing elimination of gHAT;
STROGHAT will provide better data by using state of the art diagnostics, which should enable to develop more robust models (WP2&3&4).

1.2 Methodology

Overall methodology

We will conduct an epidemiological study in an area at-risk for gHAT in DRC, incorporating a phase 3b safety study (for details see clinical study information in annex). The area was selected based on an update of the WHO HAT Atlas using cases notified in 2019, the last pre-Covid year. ‘At risk’ was defined as an estimated annual risk of infection of 1/10,000 or above resulting in 20 potential intervention areas in DRC. Among those we selected an area that is relatively easily accessible and that is surrounded by areas of low to negligible risk. Security concerns were also taken into account. Thus, we opted for the area outlined by the blue dotted line in Fig. 2 below, in the province of Nord Equateur.

The study area (Fig. 3) has 403 villages and is made up of two moderate risk clusters for gHAT. The

total population is estimated to be between 2 and 2.5 million, based on data from Landscan and WorldPop. During the first study year, all villages in the study area will be mapped and more accurate population estimates will be obtained. These estimates will be used to draw a cluster random sample to assess gHAT prevalence in the final study year.

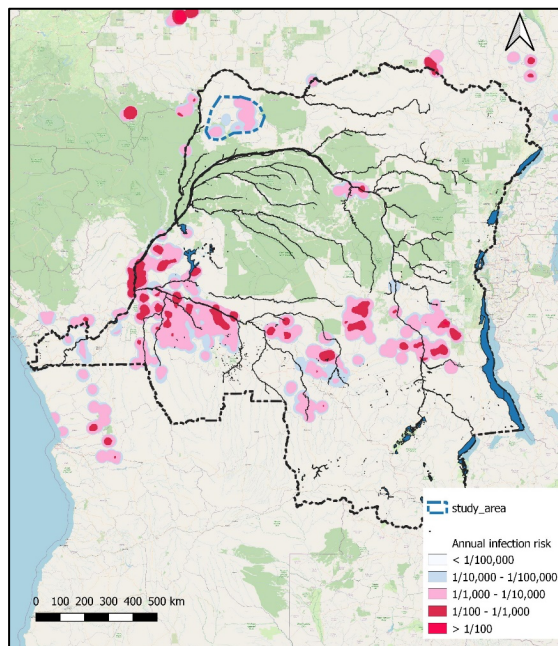


Fig. 2: Moderate to high gHAT risk clusters in DRC, based on 2019 data. (Source WHO) (intended study area inside blue dotted line)

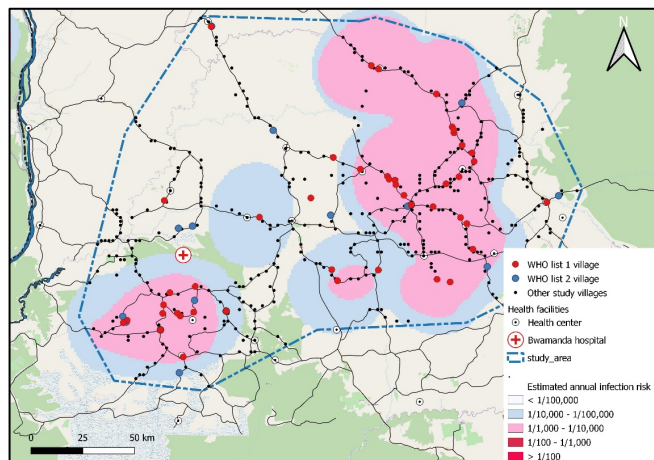


Fig. 3: The intended study site in Nord Equateur province, DRC

Within the area there are 63 villages that would need to be screened by mobile teams according to the current WHO guidelines, 48 because they had incident cases in the last three years (2018-2020, list 1 of WHO) plus 15 that had no cases in the last three years but did have cases less than five years ago (2016-2017, list 2 of WHO). The population of these 63 villages is estimated at 175,000 and will be the target population to be screened by mobile teams in the first 3 years of the study. Villages from list 2 where no further gHAT cases are detected, will be removed the next year. Villages on list 1 that had no cases for three years will be moved to list 2. No gHAT cases have been reported over the past five years in the other 340 villages, which therefore do not need active screening according to the WHO guidelines, but are covered through passive screening by fixed health centers.

Assuming that at least 85% of the population will participate and that 1% will test positive in the screening test, we expect to treat 1,500 individuals in year one and approximately 1000 more during year two and three. This complementary sample size of 2,500 participants will be added to the DNDi-OXA-04-HAT-study (900 active +300 placebo) and will provide over 90% power to detect even uncommon adverse events with a frequency of 1 per 1,000.

We will include both genders and all ages for gHAT screening. The *Screen & treat* approach will be conducted as follows:

- Information of the population by the village health worker (RECO) and head of villages to prepare the population of the screening campaign.
- A mobile team will perform door-to-door screening, using CATT for specific antibody detection.
- If eligible, the investigator will offer the seropositive individual the opportunity to participate in the study. The eligible participant will be asked to sign an informed consent.

Inclusion criteria

- Males and females of 15 years old or more
- HAT serological positive test
- Informed consent signed and validated by the investigator
(the informed consent will include permission to obtain blood sample, accepting acoziborole treatment, performance of pregnancy test, and in case of pregnancy agree to 3 monthly follow-up and clinical assistance for delivery)

Exclusion criteria

- Former HAT patient, as serology is known to remain positive for years
- Clinically significant medical condition (severe illness) that could, in the opinion of the Investigator, jeopardize the subject's safety or participation in the study
- Rejection to sign the specific informed consent form and to participate in the study

From included participants, a venous blood sample (plus lymph node aspirate if applicable) will be collected which will be shipped to the national reference laboratory for HAT for retrospective analysis to confirm gHAT. Acoziborole will be administered and the individual will be followed for 5 days by the nurse to collect adverse event. Data will be captured electronically. Three months after treatment, participants will be revisited for clinical examination and collection of adverse events. Of all participants that were confirmed as gHAT cases by molecular testing, a new venous blood sample will be collected and stored at -20°C to be tested for the presence of RNA of *T.b. gambiense*.

All individuals who do not agree or are not eligible to be included in the study will be managed according to national guidelines of gHAT. Women of childbearing age will be offered pregnancy testing prior to enrolment. If positive they will still be eligible for enrollment, but will be followed up every 3 months and to every mother it will be proposed to be hospitalized at end of pregnancy to ensure assistance at delivery. New-born exposed *in utero* will be followed during 2 years after birth.

Children below 15 years old will be included only if the results of the pediatric study (DNDI-HAT-OXA05) are available (more information to be found in the clinical study information in annex). These results will be presented to the EDCTP project management committee as well as to the project advisory committee and the DSMB of the study. Only if these external committees and the ethical committee approve the extension of treatment to seropositive children, we will include this population in STROGHAT. Otherwise children below 15 will be managed according to the national gHAT guidelines. Breastfeeding women will be included only once data concerning safety of breastfeeding while taking acoziborole will become available.

In addition, the *Screen & treat* strategy will also be offered in the 30 gHAT sentinel sites supported by WHO that already cover the area for gHAT surveillance. Instead of CATT, an RDT will be used. The decision to apply an RDT will be based on the WHO guidelines for gHAT sentinel sites. These facilities will be provided with sufficient quantities of RDTs. Anyone testing positive and residing within the study area will be offered the option of being enrolled in the study, in which case an informed consent will be signed and a venous blood sample (plus lymph node aspirate if applicable) will be collected and stored. The study team will be contacted and will ensure that the sample is forwarded to the national reference laboratory for HAT at INRB and the participant is visited at home for provision of the study drug, post-treatment observation during 5 days, and further follow up at 3 months and as appropriate.

Study design

The ideal comparison would be between an area in which standard of care is implemented and an area in which *Screen & treat* is implemented. Given the already very low incidence levels of gHAT, demonstrating a difference between such study arms would require a sample size that would not be realistically achievable. We therefore opted for a quasi-experimental design aimed at demonstrating a prevalence of below 1 per 50,000 after three years of intervention. We assume that at such levels of prevalence R_0 will be below 1 and the disease will head towards extinction. The known endemic villages at the start of the study will be screened in their entirety at the end of the study, allowing a very precise estimate of gHAT prevalence after the intervention.

During the final survey in the 4th year, villages that were originally in list 1 or 2 or were added later during the study, will be screened exhaustively, i.e. the whole population and all ages. All seropositives will be tested further with serological and molecular tests at the national reference laboratory for HAT at INRB. Prevalence of gHAT will be compared before and after implementation of the new strategy. For the purpose of the study, a gHAT case is defined as a person testing positive by a molecular test or parasitologically confirmed.

For the other villages, where no cases were reported through the health system prior to the start of the study, we will screen a cluster random sample of 150,000 individuals, deliberately over-representing those most at risk, i.e. adults.[24] All seropositives will be tested further with serological and molecular tests at the national reference laboratory for HAT. This should provide an indication of ongoing transmission in the wider study area.

Measurement and verification

For the epidemiological study, the TRYPELIM digital tool of PNLTHA will be used. This tool is used for collection of data with the following characteristics:

- Offline digital data collection through tablets (including patient registration, diagnostics results (positive and negative), treatment, geo- & demographical data);
- Quality assurance measures for diagnostics, including pictures and videos of screening and confirmation tests;
- Dashboard to visualize data, statistics and performance monitoring in the existing web based information system.

For the clinical study, it will be complemented by a separate digital data collection tool. A database check will be performed by the data manager and a quarterly report of the database status will be provided to the project manager to ensure that the data collection is done correctly in the field. For the phase 3b trial on safety of acoziborole, a meta analysis is planned to include all Adverse Events collected after acoziborole administration. The DNDI-OXA04-HAT study of seropositives will be used as yardstick.

For quality assurance of laboratory data, a comprehensive quality assurance system, including internal and external quality control will be applied.

The measurement of costing will be straightforward as all costs of the different aspects of the control will be available, including costs for diagnostics, specimen collection, human resources, investment and operating costs. The data will be used to extrapolate costs in contexts that differ from the project area, taking into account different variables.

Challenges and Feasibility

Even at times of high prevalence, gHAT has always been known for its focal distribution, also in DRC. The STROGHAT project intends to tackle an entire gHAT focus, surrounded by an area considered at very low endemicity, to avoid infections being reintroduced in the focus. This should provide an appropriate context to demonstrate the effect of the study on the transmission of disease, i.e. the first objective. Proving zero prevalence after three years of a *Screen & treat* intervention in the targeted area will require a large sample size but this is feasible.

The study makes use of the existing PNLTHA organization and the experience of its staff. Until recently in DRC there were so many villages to be screened according to the WHO guidelines, that the available screening capacity was largely insufficient to lead to interruption of transmission in the identified foci. With only 421 cases reported in 2021, this is no longer the case. [7]

The mobile teams of the national HAT control program (PNLTHA) have extended experience to ensure appropriate mass screening, using highly sensitive and specific gHAT screening tests. Routine data from DRC show specificities of 99.0 to 99.5% for rapid diagnostic tests (RDT) and Card Agglutination Test (CATT) respectively, allowing to narrow down the population to be treated to fairly small numbers. One mobile team has a capacity to screen 60,000 individuals per year, and even during

the COVID-19 pandemic in 2020 and 2021, PNLTHA screened two million people annually. We aim at a total sample size of approximately 150,000 to be screened and 2,000 to 3,000 seropositives to be treated, by 3 mobile teams. These existing teams will be expanded by 1 medical doctor and 1 nurse, and additional means of transportation for the time of the clinical study, to ensure that additional tasks inherent to the clinical trial, can be ensured.

In addition to mobile teams, also a number of **fixed health facilities will be included in the study**. These facilities will be mainly selected from the 30 WHO sentinel sites that already conduct gHAT activities in the area according to WHO guidelines. The additional workload of the national reference laboratory to implement tests has been assessed (3.2. WP2) and additional staff has been budgeted.

The number of seropositives to be treated is higher than numbers of HAT cases currently treated in DRC but since this is a single dose oral treatment, it will be logistically much easier than all other treatments previously used. Acoziborole will be provided through DNDi, an associated partner.

Complementary to diagnosis and treatment, vector control also takes part in gHAT control. Over the last decade, major advances have been made through the introduction of “Tiny Targets”. This new device has potential to reduce the vector density in a systematic way and accelerate elimination.[19-20] However, in STROGHAT we will limit ourselves to a **'Screen & treat'** in order to clearly show the effectiveness of this approach, and avoid potential bias from vector control implementation.

Interaction with other research and innovation activities

The STROGHAT study will build upon recent or ongoing research and innovations in gHAT control, in which STROGHAT partners are or were involved. The partners involved in the study all have already proven to be *fit for purpose* for the work that will be expected (see 3.2). They all have extended experience in conducting clinical trials, diagnosis and treatment of HAT, and managing HAT control.

ITM, IRD, DNDi, PNLTHA and INRB have successfully participated and collaborated in several projects, including but not limited to European projects under FP7, Horizon 2020 and EDCTP2 aiming at improved diagnosis of HAT:

- **TryLeiDiag** (FP6-INCO ID: 15379; 2005-2009, coordinator ITM partner INRB) aimed to develop and validate point-of-care tests for molecular leishmaniasis and HAT diagnosis based on ribosomal RNA detection.
- **Nidiag** (FP7-HEALTH ID: 260260; 2010-2016, coordinator ITM, partner INRB) aimed to develop simple, cost-effective diagnosis-treatment algorithms for three neglected infectious disease related clinical syndromes: the persistent fever, the neurological and the digestive syndromes. The first commercial RDT for HAT, HAT Sero-K-Set was developed in this project.
- **DiTECT-HAT** (EDCTP2, Horizon 2020, DRIA2014-306-DiTECT-HAT, 2016-2021 coordinator IRD, partners ITM, PNLTHA, INRB and collaborating with DNDi) aimed at delivering new, cost-effective diagnostic algorithms for gHAT elimination. Diagnostic accuracy of RDTs for HAT was evaluated, and detection of RNA was confirmed as a powerful tool for diagnosis of HAT and detection of cure [17]
- **SpeSerTryp** (FIND funding, partner IRD, 2022) evaluates **specificity of all HAT screening tests** (CATT and 4 RDTs) currently under development or available in Guinea and Côte d'Ivoire. A similar study is ongoing in DRC, conducted by ITM and PNLTHA, in the framework of the Belgian supported “HAT+” project. The results of these studies will be exploited for choosing the best test in STROGHAT.

The participants are also participating in several ongoing EDCTP2 projects on treatment of HAT:

- **ACOZI-kids** (EDCTP2, Horizon 2020, RIA2019PD-2890-ACOZI-KIDS, 2020-2025, coordinator DNDi, IRD partner): pediatric study (DNDI-HAT-OXA05) assessing safety and tolerability of acoziborole in children in DRC and Guinea.
- **Fexi-g-HAT** (EDCTP2, Horizon 2020, CSA2018HS-2526-FEX-g-HAT. 2020-2024, coordinator DNDi, partners IRD, ITM and PNLTHA) is facilitating the delivery and uptake of fexinidazole, the first all-oral treatment for HAT, into selected African endemic countries through strengthened health care capacity and coordination. This network will also facilitate uptake of STROGHAT results

Furthermore, DNDi, PNLTHA and INRB already jointly conducted the pivotal phase 3 trial (DNDI-OXA02-HAT) on acoziborole revealing effectiveness and safety in gHAT patients.

As the gHAT community is limited, all project partners have good connections with other gHAT research and control stakeholders that are not directly involved in STROGHAT. This should facilitate taking into account research and innovations from such expert groups. STROGHAT intends a strong interaction with WHO and compliance with the output of different WHO expert committees (i.e. for diagnosis and treatment), in which several STROGHAT partners also take part. All partners collaborate with the HAT Modelling and Economic Predictions for Policy (HAT MEPP) supported by BMGF (INV-005121). STROGHAT will ensure that HAT MEPP can use the new data, which should lead to improved models.

STROGHAT will make use of innovations already introduced in the HAT control program in DRC such as the TRYPELIM digital and project management. In recent years DNDi has invested in the implementation of gHAT awareness raising campaigns, via a project financed by the World Bank. STROGHAT will build further on this experience.

Four year follow-up of STROGHAT

STROGHAT will closely interact with interventions in support of gHAT control, whereby the most important funding is currently ensured by BMGF, the Belgian and French Government, Sanofi and Bayer.[21] These donors will be regularly informed about intermediate results during the implementation of STROGHAT.

BMGF and the Belgian Government have already committed to continue this support until 2030, and are very interested in the results of STROGHAT (BMGF already contributes in kind). Partners ITM, PNLTHA, INRB and IRD are main actors in the implementation of the BMGF projects TRYPELIM (INV-031337) and Trypa-NO! (INV-001785) in support of elimination of gHAT in respectively DRC, Côte d'Ivoire, Guinea, Chad, Angola and Uganda. ITM coordinates the efforts of the Belgian Government in DRC (HAT+ and FA5). The results of STROGHAT will influence the strategic orientations of the next phase of all these projects.

The results of STROGHAT will also enable to reassess the strategy of vector control in gHAT control, which is part of TRYPELIM and Trypa-NO!. However, as it is not feasible to deploy the devices everywhere, a more targeted approach is needed.

Interdisciplinary approach

As will be clear from the above, STROGHAT brings together expertise and methods from different disciplines in the pursuit of its objectives. The consortium brings together researchers and implementers such as PNLTHA but also the primary health system, involved via the passive screening. Among the researchers there is a wide variety of expertise including clinical medicine, epidemiology, parasitology, health policy, health economy, social sciences, information technology, clinical trial organization, pharmacovigilance with ample experience in implementation research. In addition also the end users of the results of STROGHAT are represented in the consortium.

Each discipline has a clearly defined role in preparing the ground for the epidemiological study, the clinical trial and the costing analysis

Role of social sciences

A high participation of the selected villages is important to successfully bring down HAT prevalence, which explains that a lot of effort will be invested in raising awareness. Since decades, public awareness campaigns making use of village health workers are part of the PNLTHA strategy to increase participation in active screening. These so called *Relais Communautaires (RECOs)* are elected by the villagers and facilitate interactions between the health system and the villages. These RECOs will be involved throughout the study, before the arrival of mobile teams up to follow-up of treated patients.

Before starting the trial an acceptability study will be conducted by social scientists, to explore

acceptability of the study procedures and factors associated with participation. Screening activities at village level will be preceded by an information-education-communication campaign to inform the population on the changes in screening procedures and treatment. The study population is already familiar with the yearly passages of the mobile teams. It is expected that omission of lumbar puncture from the protocol and the provision of a single dose treatment on the spot, without need of hospitalization, will be important assets to maximize participation.

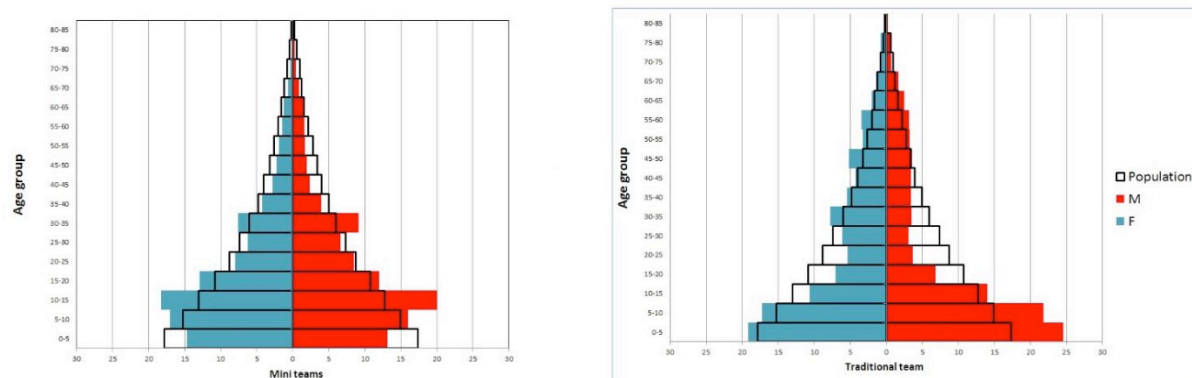


Figure 4: Age and gender distribution of population screened by mini and traditional mobile teams in Bandundu in 2018 (DRC)

Gender dimension

It is intended to include children and women, including pregnant women, into the study and the preparatory work to increase participation will certainly focus on these groups. As described above, all women of childbearing age will be offered pregnancy testing prior to enrolment, to complete the safety and efficacy data, but not to exclude them from the study.

Traditionally women and children have been an major target group of the HAT control program in DRC, but an important concern is the participation of young adult men. An analysis of routine data from DRC between 2006-2008 showed an overall slightly larger proportion of men among cases diagnosed (52.5%) but there were differences between provinces. In Bandundu the majority were women (56.1%) whereas in Kasai the majority were men (59.9%). One in seven patients in Kasai was working in the diamond mines, exclusively men. These diamond mines are all situated along rivers, corresponding with the habitat of tsetse flies. What these data show is that gender does play a role but that there may be important regional differences related to different levels of exposure to tsetse flies. More recently, we explored age and gender distribution among subjects screened by traditional mobile screening teams that gather village populations in a central location and mini teams that go door-to-door, all in the former Bandundu province of DRC. Figure 4 (right-hand side) shows how young adult men are particularly underrepresented in active screening by traditional teams which could mask gender differences in infection risk. As shown in Figure 4 (left-hand side) the underrepresentation of young adult men was far less apparent among populations screened through door-to-door screening.

The most probable explanation is that adult men are reluctant to participate in HAT screening as this is perceived as time-consuming. Door-to-door screening takes less time for those screened and can be planned at more convenient hours. In the STROGHAT study, we will therefore conduct door-to-door screening and record all inhabitants of each household visited.

Thus, we will be able to assess gender and age distribution among participants recorded and verify that all groups are well represented. We will then verify whether the age and gender distribution of subjects screened matches that of subjects recorded to ensure that both genders and all age groups are adequately covered.

Methodology with respect to EU Taxonomy regulation

The products and the activities provided in the framework of the study are considered to have no environmental impact. Vector control, which could potentially have an impact on biodiversity, is part of the gHAT control strategy in DRC, but will not be part of this study. Hence the methodology is designed in a way that will not significantly harm any of the six environmental objectives of the EU Taxonomy regulation.

Open science

Open science practices will be implemented as an integral part of the proposed methodology. As part of the consortium agreement, we will develop a data sharing policy, maximizing the early sharing of research outputs and study metadata, such as study protocols (through study registration in clinicaltrials.gov), data management, statistical analysis plans, and developed tools (e.g. through the project website or public registry). This policy will be in line with the applicable European and local regulations. Since the proposed project will be conducted as a GCP-compliant clinical trial, collected participant-level data can only be shared before study completion for well-defined purposes, such as safety monitoring, follow-up of populations screened by national HAT control programs, and predefined interim analyses.

An open-format annotated study database (such as .csv files) from which personal identifiers and other sensitive data have been removed, will be made open access through ITM's research portal as well as the DNDi portal, suitable public repositories with persistent identifiers for data sharing, the project website for policymakers as well as the scientific community. Datasets including personal data can be made accessible for secondary research on conditions of data anonymization and aggregation, subject's consent, a Data Access Committee request review and approval procedure, and a Data Sharing Agreement, in compliance with privacy regulations such as the EU GDPR.

The key outcomes will be made publicly available within 12 months of study completion by posting the results section of the primary clinical trial registry (e.g., www.clinicaltrials.gov). Likewise, shared ownership of the data between the project participants will be explicitly described in the consortium agreement and the data management plan. Scientific publications originating from the project will be open access.

PNLTHA has a mandate to organize scientific consultative meetings on gHAT in DRC, where new insights, results of research and new tools are presented and discussed. The participants in such events include not only Congolese and international scientists specialized in gHAT, but also representatives from other directions of the Ministry of Health (e.g. primary health care system), academics and the civil society that are not directly involved in gHAT. It is obvious that the (intermediate) results of this project will be discussed at this event.

Research data management and management of other research outputs

Types of data/research outputs: The quantitative survey will generate epidemiological, clinical and laboratory data. Data of the clinical study will be captured and processed in the open access and Good Clinical Practice compliant software system MEDRIO used by DNDi, assuring integrity, accessibility, and confidentiality of the data. Epidemiological data will be collected using the TRYPELIM tool, routinely applied by PNLTHA. The costing data will be obtained from the management of the project and from PNLTHA, INRB and ITM.

Findability of data/research outputs: Research documentation and outputs will be made visible and accessible as soon as possible. As this is a clinical trial, it will be prospectively registered in a WHO-accepted primary registry, before any participant inclusion, including details on investigators, milestones, sites, research design & methodology, etc. Study results will be uploaded upon completion of *ex-ante* analysis. Curated research datasets, metadata and other essential supporting documentation will be made available through public repositories (e.g. ZENODO) and via an existing website managed by ITM. Wherever possible, persistent identifiers will be attached to research outputs.

Accessibility of data/research outputs: Given that we will process sensitive categories of personal

data (medical data), the EU GDPR applies which prohibits the full open access to such data. Measures such as data minimization, anonymization, review and approval by Data Access Committee and a Data Sharing Agreement will be put in place to assure confidential secondary use of personal data. These data will be made findable by means of data access statements in scientific publications, publication of metadata about the datasets on the project website and persistent identifiers linked to the datasets. Consortium members will agree on controllership over the data and set up a personal data sharing policy. Considering that the project deals with large-scale processing of sensitive data, we will also perform a Data Protection Impact Analysis as required by the GDPR. Mathematical model code and results will be made publicly available through scientific publications but will not contain any personally identifiable data.

Interoperability of data/research outputs: The project will comply with the ICH Good Clinical Practice (E6-R2) guidelines including data management, thus assuring traceability, integrity, and confidentiality of the data. Where applicable, clinical coding will be done per the MedDRA standard.

Reusability of data/research outputs: Ownership of research outputs will be fully shared between the consortium research partners. Moreover, we will adapt open licenses for all open sharing of outputs and data, except where it concerns sensitive data. The data collection system developed in MEDRIO will towards the end of the project be adjusted for routine use by national HAT control programs. It will be made available to any HAT control program that wishes to make use of it. Detailed instruction manuals, and SOPs will be developed for this purpose and made freely available.

Curation and storage/preservation costs: According to legislation and partner institutions archiving policy, clinical trial essential documentation will be retained for 25 years. The ITM will be responsible for securing long-term storage, and PNLTHA will assure secure storage on-site, in accordance with local legislation.

2. Impact

2.1 Project's pathways towards impact

As shown in Fig.1, the current control strategy has proven effective to rapidly reduce the gHAT prevalence, but there are questions whether it is feasible to reach full interruption of gHAT transmission. The current strategy of active and passive screening in DRC is not adapted to further progress rapidly within an epidemiological context of low and very low prevalence. The integration in the primary health system is not an effective alternative as its capacity is limited. In addition, sustainability of HAT control in DRC risks to be threatened, not only because it depends on continuation of external funding, but also because the DRC faces many far more important public health challenges than gHAT.

The arrival of the acoziborole, an oral single dose and safe drug is a game changer and the *Screen & treat* strategy proposed by STROGHAT provides several advantages over the current strategy, including:

- Less laboratory workload in the field due to skipping microscopic examinations
- Larger coverage of human parasite carriers
- Improved compliance by eliminating the need for lumbar puncture or hospitalization
- More suitable for integration in the health system

In turn, facilitated diagnosis, improved acceptability and wider access to treatment will lead faster to interruption of transmission.

However, a number of prerequisites need to be fulfilled before a new strategy for interruption of gHAT transmission can be recommended by WHO and internationally accepted and implemented by health

decision makers in affected gHAT countries. In line with its primary objective and the two secondary objectives, STROGHAT wants to play a major role to facilitate this process.

Expected results

The implementation of STROGHAT will generate 3 major results:

a) Evidence of effectiveness of the *Screen & treat* approach

This result will be based on 2 indicators. 1° the comparison of prevalence before and after implementation of *Screen & treat* in 63 endemic villages; and 2° whether or not there is indication of interruption of transmission of *T.b. gambiense* in the focus, based on data from a cluster random sample in the wider study area.

The first indicator will determine if zero transmission has been reached in the endemic population of 175,000 in the villages where the *Screen & treat* strategy was applied during 3 consecutive years. For the other villages, where no cases were reported through the health system prior and during the study, a cluster random sample of 150,000, deliberately over-representing those most at risk will provide an indication of ongoing transmission in the focus with an estimated total population of 2.5 million.

Enablers to reach the expected result:

- The strategy will be conducted by teams that have extended screening experience in the study area and that will be re-enforced during the study
- Results of other ongoing studies, expected in the coming months, will allow inclusion of more individuals, in particular children
- Participation will be maximized by awareness raising campaigns
- Known barriers to participation to screening have already been dealt with (e.g. obviating lumbar puncture)
- Provision of free treatment on the spot will encourage the population to participate in screening activities by mobile teams in selected villages. Provision of free, single dose acoziborole will also motivate patients outside the selected villages to visit health facilities or have themselves tested during screening in a close by village.

b) Additional information on safety of acoziborole

At a WHO meeting on “Integration of new tools into national and global policies” (11/12/2019), a Target Safety Profile for escalating extension of use of a single-dose oral medicine was agreed upon that is needed to justify the adoption of a *Screen & treat* strategy. The needs identified were

1. Larger studies enabling to detect uncommon adverse events (AE) (1/100 to 1/1,000)
2. Safety data in pregnancy
3. Safe in children (ideally all ages, but at least in >5 years)

STROGHAT will make a major contribution to the first aspect, by generating safety data involving an estimated 2,500 individuals treated in their village. Other clinical studies have been conducted or are ongoing to collect uncommon adverse events and safety data in pregnancy and children (see higher).

Enablers to reach the expected result:

- Several clinical trials on NECT, fexinidazole and acoziborole have been conducted in DRC in previous years by DNDi and PNLTHA in several other provinces and sites (former Bandundu and Kasai provinces). This experience will facilitate the implementation of this new clinical study, by the same teams, in another province, Nord Equateur.

c). Costing of implementing a *Screen & treat* strategy at population level

A comprehensive insight in costing of a *Screen & treat* approach at population level is essential to develop a realistic long-term plan to reach the elimination of disease and is acknowledged as a critical action point for all neglected tropical diseases. [21] The focus will not be on the cost per infected person or on cost effectiveness, because it is obvious that cost per infected individual is bound to continually raise as the elimination program progresses and approaches its goal of zero transmission.

Preparation of a costing tool is a first step towards adoption of the strategy in other foci, in and outside the DRC.

Enablers:

- The collection of these costs is straightforward, as these data need to be collected for the project. Partners ITM and PNLTHA have previously performed costing analyses together.[22]

Outcome

Over the medium term, the expected results will contribute to 3 outcomes:

a) Official recommendations and guidelines by WHO concerning the use of acoziborole

Based on the results of the clinical trials, acoziborole will be submitted for opinion to the European Medicines Agency (EMA) under Article 58 (human medicine intended exclusively for markets outside of the European Union). The submission to EMA is planned Q4 2023 (positive opinion expected Q1/Q2 2024), followed by submission in DRC for marketing authorization. As was previously the case for fexinidazole, [23] once EMA establishes a positive opinion, WHO will soon after convene its Guideline Development Group (GDG) for HAT treatment which will formulate new or updated guidelines for the use of acoziborole.

Enablers:

- Several conditions for adopting wider use of acoziborole have been already defined by WHO, including a “Target Product Profile for a HAT medicine to treat individuals without confirmed infection”, and a “Target Safety Profile for escalating extension of use of a single-dose oral medicine” (WHO meeting on “Integration of new tools into national and global policies”, 11/12/2019). In 2022, WHO has also already published a target product profile for a gHAT test to identify individuals to receive widened treatment (ISBN: 978 92 4 004329 9, <https://www.who.int/publications/i/item/9789240043299>). Proactive existence of these documents, shows the importance WHO attaches to development of new *Screen & treat* strategies as proposed by STROGHAT. The strong involvement of WHO is also illustrated by acceptance of WHO officers to participate in the STROGHAT external scientific advisory board, would the project be funded. These factors should facilitate the process of adoption of acoziborole.

b) Acceptance and roll-out of *Screen & treat* as state-of-the-art strategy in both passive and active screening by national control programs of gHAT endemic countries.

Once STROGHAT results have been communicated and in particular once recommended by WHO, national HAT control programs will progressively put in place *Screen & treat*.

Enablers:

- WHO organizes annual stakeholder meetings uniting directors and the main officials of the national HAT control programs of all endemic countries
- Members of national HAT control programs are also informed through regular meetings and newsletters of the HAT platform
- STROGHAT project partners are supporting national programs of DRC, Angola, Côte d’Ivoire, Guinea and Uganda through the TrypElim and Trypa-NO! projects.

c) Acceptance of *Screen & treat* as state-of-the-art strategy in both passive and active screening in HAT foci by local health workers and the population at risk.

The advantages of *Screen & treat* -facilitated diagnosis, improved acceptability and wider access to treatment- over the current strategy for both caregivers and caretakers are important.

Enablers:

- The number of individuals screened in STROGHAT is important which should facilitate acceptance of results
- The use of the most sensitive diagnostic techniques available is an asset

- The inclusion of a quality assurance system, involving the national reference laboratory will enable decision making, and control use of acoziborole

d) Long-term funding plan to progress to elimination of disease in all affected countries

Within STROGHAT, a tool will be developed to facilitate cost estimations in other areas of DRC and in other affected countries which will enable national control programs and their funders to estimate costs related to *Screen & treat* in their local context.

Enablers:

- The Bill and Melinda Gates foundation already finances control of gHAT in DRC, Guinea, Côte d'Ivoire, Chad, Uganda and Angola. Their interest in *Screen & treat* is shown by the support letter provided for the STROGHAT proposal
- The Belgian Government is a long-term funder of the control program in DRC and signed a MoU with BMGF to jointly take the lead in supporting the gHAT elimination agenda.

Impact

As already suggested above, the longer-term effects will have an impact at different levels

Individual

A major impact will be the improved access of infected individuals to quality gHAT treatment, without a necessary visit to a health facility. Despite the introduction of less toxic drugs since 2009 (NECT and fexinidazole) access to quality treatment has remained challenging. Because of the 10-day intravenous administration, NECT requires hospitalization and a minimum of capacity in a health facility. Fexinidazole can be administered to outpatients, but 10-day intake under supervision (required until further notice) necessitate a minimum of management and compliance. Furthermore, its lower effectiveness in severe late stage gHAT patients continues to necessitate a lumbar puncture, in particular in a large fraction of patients seeking care through passive screening.

Direct and opportunity costs for travel and potential hospitalization will be considerably reduced. In addition to improved care, treatment access will be improved, in particular for patients with low parasite loads in blood in whom presence of the parasite is difficult to demonstrate. In the best case, such patients are actually followed-up and reexamined annually, –at least if the control program has the means to do so-. Many of such patients undergo follow-up for years before being confirmed, but a majority gets discouraged and drops out with the risk of dying of gHAT. *Screen & treat* implementation will thus offer quality care to infected individuals that are currently not treated.

Organizational

A *Screen & treat* implementation will enable to increase effectiveness of PNLTHA, because it will enable to:

- Hit the remaining gHAT transmission foci early and hard, which in turn, will speed up the epidemiological decline of gHAT in the entire continent.
- Increase screening coverage in an effective way (e.g. difficult accessible areas).
- Rapidly react to resurgence in an effective way.
- Increase quality of gHAT diagnosis in both active and passive screening.

A *Screen & treat* strategy will also enable to review the organization and rationalize gHAT elimination, because specialized staff and equipment needed for confirmation techniques on the field can be reduced. As a consequence, more people can be screened per day and treatment administered immediately, which will impact effectiveness and efficiency.

A considerable impact is expected in logistics. Combining screening and treatment means that diagnostics and treatment do no longer follow 2 different chains, which currently is considered a major challenge leading to non-treatment of a considerable number of confirmed individuals.

Institutional

The implementation of a *Screen & treat* strategy including a posteriori diagnosis in a national reference laboratory, will need a reorganization and review of the tasks and mandates of PNLTHA and INRB. Also, the integration of gHAT control in the primary health system should be thoroughly reviewed.

The STROGHAT study will not only be relevant for DRC but will also be relevant for health authorities of all 25 sub-Saharan African countries (previously) endemic for gHAT, once WHO recommends a *Screen & treat* strategy.

Wider impact

The wider impact will be immediately linked to reaching the overall objective, i.e **Obtaining interruption of gHAT transmission**.

Obtaining the objective will be an enormous success with impact at different levels:

- Scientific: elimination of a human parasitic disease has not been reached before. Elimination of smallpox was based on vaccination, a tool that is not available for gHAT. The impact of elimination will definitively give a big boost to the scientific society in general and the Neglected Tropical Diseases (NTD) in particular, as it would prove that investment in research and innovation can provide a considerable return on investment.
- Economic: gHAT has had a considerable economic impact on the population of endemic countries (e.g., agriculture and fisheries). In addition, gHAT affects the rural poor in sub-Saharan Africa, where economic success stories are scarce. Removing a lethal disease will be an important asset for improved health and thus productivity
- Societal: the societal impact of gHAT on a community cannot be underestimated, not only at individual and household level, but also on the community at large. Despite the current low prevalence, the disease and its effects are well known and feared, in particular by the older population.

Scale and significance

Interruption of transmission can only be reached if all areas at risk will be tackled. As already mentioned, gHAT has a very focal distribution and previous outbreaks tend to reemerge from the same historic foci. Therefore, it will not be necessary to expand the *Screen & treat* strategy across all 70 million estimated to be at risk, but the current screening strategies do not enable to guarantee an effective coverage, in particular in poorly accessible regions.

The *Screen & treat* coverage can prove an asset to solve this problem, in all affected countries. The success of gHAT control in recent years cannot be denied. Currently only 11 countries still report gHAT cases with DRC still reporting the majority of cases (around 70%). However, there is a number of countries surrounding DRC, with comparable challenging contexts, and where gHAT surveillance is poor including the Central African Republic, the Republic of the Congo (Brazzaville), South Sudan, Nigeria and Sierra Leone. The *Screen & treat* strategy applied in STROGHAT is expected to provide new prospects for such countries, as it can be conducted fast and effectively by less specialized teams and probably at lower cost.

Remote confirmation with more accurate tests than those currently available in the routine control program will improve insight of the epidemiological situation. It cannot be expected that surrounding countries will be able to develop a laboratory with equal capacity as the national reference laboratory at INRB. However, it should be feasible to organize a system whereby surrounding countries that apply a *Screen & treat* strategy call upon the expertise of not only INRB, IRD & ITM but also CIRDES (Bobo-Dioulasso, Burkina Faso) and IPR (Bouaké, Côte d'Ivoire), which also have proven record on gHAT diagnostic testing, to confirm diagnosis using higher sensitivity tests.

Requirements and potential barriers

Despite the important progress in the process to test acoziborole, the drug will not be available before 2025 because a number of steps still have to be taken (e.g. EMA assessment, inclusion in the list of essential drugs of WHO, registration in DRC). However, this should not be an excuse to wait with launching the STROGHAT study. Considering the rapid epidemiological decline, gHAT field research becomes more and more difficult as well as obtaining significant results. This explains why the STROGHAT initiative is considered timely by both WHO and donors.

To deliver the outcomes, results of STROGHAT will have to be presented to and accepted by different stakeholders. WHO will need the data to adopt the international guidelines. In DRC, review and modification of the gHAT strategy is based on evaluation of results of the existing strategy and recommendations made by PNLTHA. To this effect, consultative scientific committees are organized involving not only national and international gHAT experts but also health experts in a wider scope. Presentation of results to this committee is a first essential step.

An important drawback of the STROGHAT is the focus on one particular study area in one country, the DRC. This is on the one hand due to the limited number of countries where gHAT is endemic that are member of the EDCTP association, and, on the other hand, deliberate, to focus on one geographical area so not to dilute efforts in an attempt to cover more/too dispersed areas. However, BMGF has already shown interest to compensate this gap and finance a similar study in Guinea Conakry.

A concern of applying a *Screen and treat* strategy, is the consequent massive overtreatment with a potential lower benefit /risk ratio in comparison with the current strategy. The increased safety data should provide insight in uncommon risk events but will not be sufficient to exclude rare adverse events. Considering current positivity rates in the gHAT control program of DRC, overtreatment would be limited to maximum 1% of the population considered at risk. In addition, WHO has already established a target product profile for a gHAT test to identify individuals to receive widened treatment (ISBN: 978 92 4 004329 9, <https://www.who.int/publications/i/item/9789240043299>), which focusses on high specificity.

It is known that even in areas without HAT, a number of individuals reacts positive on a gHAT screening test. This means that we need a strategy to decide when to stop screening. The application of better serological and molecular tests for all seropositives in a reference laboratory, as proposed in the STROGHAT study, will largely take care of that concern. Accurate identification of true positives, even at a later stage, is an important asset for better decision making of operational issues and is the topic of another WHO Target Product Profile proposed by the WHO HAT Diagnostic Technical Advisory Group, which is awaiting public consideration.

Operational issues are by far the main challenge to reach the expected results and outcomes in DRC. Critical risks are presented in Section 3 (see further) but the list is endless due to a complex context (e.g. accessibility, security, instability, ...). The complexity is aggravated by operational challenges outside DRC (e.g. unavailability of diagnostics in recent years, shipment delays, ..), explaining why STROGHAT invests in administrative and logistic support, which will facilitate to solve operational issues. It should also be noted that progress in HAT elimination of the last years has been reached with more complex tools and despite the difficult context.

A major asset of the STROGHAT consortium is that it brings together the government HAT control program and the national reference laboratory of the most endemic country and most of the key international partners, directly or indirectly involved (the latter will be regularly updated).

A potential barrier is lack of interest from manufacturers to produce tests that have very limited market value because the disease may be about to disappear. The production of gHAT RDTs has been problematic, although following a long-term contract for the production of HAT Sero-K-Set this issue seems to be solved. Notwithstanding, this issue requires close follow-up in order to timely react. ITM already committed to produce CATT for as long as it takes.

In comparison to other gHAT affected countries, DRC has relatively important resources and it may not be evident for surrounding countries to implement the system tried out in STROGHAT, in

particular in relation to *a posteriori* testing at centralized laboratories. INRB, ITM and IRD are all 3 recognized as WHO Collaborating centers for gHAT diagnosis. All three have been involved in the development of all new diagnostic gHAT tools in recent years and are continuing to develop new diagnostic tools. Through their involvement in the current project, they are in an excellent position to support other gHAT affected countries to find appropriate solutions to implement a similar system.

Target groups and beneficiaries

STROGHAT focuses on 3 target groups

- **Health authorities of DRC and other gHAT endemic countries**
Ownership of the elimination agenda by the health authorities is key to reach the final objective and obtain the wider impact. The expected outcome can only be reached if acoziborole is registered and the results of STROGHAT lead to adoption of *Screen & treat* as part of the national gHAT strategies.
- **WHO**
The adoption of the *Screen & treat* strategy in the national gHAT control strategies will only be realized if WHO also adopts this strategy
- **Donor community**
The donor community needs to be convinced of the effectiveness of the *Screen & treat* strategy, to make long-term commitments for a disease that is no longer a major public health problem in affected countries.

The direct beneficiaries will be the population of the study area during the implementation of STROGHAT (2.5 million people in the focus of Nord Equateur). The indirect beneficiaries will be the entire risk population of all gHAT affected countries (70 million).

Coherence

The objectives of STROGHAT are in line with the gHAT targets of WHO road map for Neglected Tropical Diseases 2021–2030, i.e. interruption of gHAT transmission by 2030, defined as zero cases reported.[21] Support to the elimination of gHAT is also in line with the Sustainable Development Goals (SDG) to end poverty and inequality.[24] It supports SDG 3 “Ensure healthy lives and promote well-being for all at all ages” and focuses on target 3.3 “Infectious diseases”, including the combat of neglected diseases. The project also adheres to Goal 1. “End Poverty in all its forms everywhere”, because gHAT not only often affects the rural poor in difficult accessible areas, but it also tends to increase poverty of vulnerable affected households.

2.2 Measures to maximize impact - Dissemination, exploitation and communication

Scientific Community

All partners of STROGHAT have a tradition of reporting results in open access scientific journals. At the start of the project a publication plan will be agreed upon between all consortium partners. We foresee scientific publications (at least one per year) in relevant peer reviewed open access platforms.

We will also participate in the ‘**International Scientific Council for Trypanosomiasis Research and Control**’ (ISCTRC) which is a biennial scientific conference organized by the African Union. Other relevant international tropical disease conferences in which the STROGHAT consortium intends to take part, are the ‘Coalition for Operational Research on NTDs’ (COR-NTD), and the American Society of Tropical Medicine and Hygiene (ASTMH) meetings which traditionally take place in

November, and the two yearly European Congress on Tropical Medicine and International Health (ECTMIH). The World Health Summit could be an additional high-level meeting to be targeted, and in which a session of HAT took already place in 2019. The different partners of STROGHAT have a tradition of organizing sessions in these fora for several years. IRD, ITM and INRB also have a tradition to organize fora focused on francophone Africa, to enable scientists that do not master English to also take stock of progress.

WHO and Policy level

WHO is expected to play a significant role in the dissemination of the results of the STROGHAT project. The WHO plays an important role in the control of gHAT, not only in developing appropriate strategies and guidelines, but also in ensuring coordination between affected countries and partners. They organize expert committees and consultative groups (e.g. HAT Elimination and Diagnostic Technical Advisory Groups, Development of new tools and Integration in national policies, ...) of which several members of the consortium are part. WHO facilitates the organization of training and the delivery of gHAT drugs and raw materials for HAT diagnostics, donated by the pharmaceutical industry. WHO experts have actively contributed to developing the proposal and have agreed to be on the scientific advisory board.

For dissemination at policy level, we will make use of the ‘Network for HAT elimination’ established under the auspices of WHO that organizes biannual meetings of all stakeholders as well as annual country coordinating meetings.[25] All members of the STROGHAT study consortium are represented in this forum and different expert committees mentioned above.

In addition, PNLTHA, WHO and ITM jointly organize partner meetings for gHAT partners active in DRC in which program implementation and progress towards HAT elimination are discussed by all partners of the PNLTHA. Consortium partner DNDi hosts the regional ‘HAT Platform’ that organizes annual scientific meetings in HAT endemic countries.[26]

The DRC policy includes organizing *Comité Consultatif Scientifique* meetings by PLNTHA. At such occasions, international and national experts with diverse expertise backgrounds gather and discuss innovations and results. The committee can recommend modifications of the National control strategies, that are submitted to the Ministry of Health. The outcomes are the result of consensus, also taking into account opinions from experts that are not directly involved in HAT. The advantage is ownership on a broader level, but it explains why progress is slow. In the past, policy decisions taken by DRC have paved the way for other countries to follow suit (e.g. registration of fexinidazole). It is trusted that the results of STROGHAT will have an impact on all gHAT strategies.

Political

Ownership of the political leaders of the affected countries is considered crucial for the endgame of elimination. All partners directly involved in STROGHAT have a tradition of continuously putting gHAT on the agenda of the political gatherings at all levels. Heads of States and high-level health authorities have been sensitized over the last decade, not only at prominent NTD fora but also at political fora such as the “*Sommet de la francophonie*” (francophone summit), attended by political leaders of affected countries. Several political leaders have already noticed that gHAT control and elimination have so far been a success story. The results of STROGHAT will be a good occasion to renew interest.

Also, EU would benefit from its high visibility, leadership and standing, especially in partnership with Africa. Already EU member states Belgium and France are making major contributions in the fight against gHAT. The STROGHAT study could become an important catalyzer in highlighting the role of EU member states in gHAT elimination and pave the way for other successes.

Financial actors

All important donors of gHAT control have already been informed about the intention to submit the current study proposal and are eager to closely follow-up the obtained results, which could have an impact for their future commitments.

Public at large

PNLTHA has a tradition to organize an annual national gHAT day at the end of January, including a press release, and involvement of politicians. It is obvious that the results of the STROGHAT study will be disseminated at this event, which will enable dissemination to a wider audience.

Advocacy among the targeted population will be an integral part of the study strategy, as maximal participation is essential to obtain the expected results. However, given the low prevalence of gHAT already reached, such Information-Education-Communication (IEC) campaigns need to be carefully targeted to avoid unnecessary costs and enable replicability. The STROGHAT project will develop IEC materials, in such a way that it should also be useful in a wider context of sub-Saharan Africa. PNLTHA, DNDi and ITM have already built-up experience in this domain in DRC (e.g., IEC in framework of introduction of fexinidazole via PDSS, a World Bank project in DRC), while IRD has extended experience in this domain in Côte d'Ivoire and Guinea Conakry. This will allow use of the communication material in all HAT endemic countries after the project.

All partners have good relationships with the press and have dedicated services for science communication, which will enable the project to communicate to a larger public (e.g. IRD Le Mag', The Conversation etc.). Publications appearing in "The Conversation" reach a public of millions of science interested readers, and are often taken over by the general public quality journals (cfr. an article on the EU funded COMBAT project on control of animal trypanosomosis in El Pais on 1/03/2022). Last but not least, the project will have an active website and Twitter account.

Research Capacity in Sub-saharan Africa

The STROGHAT project contains a strong training component, in particular on all aspects of clinical trial implementation and on diagnosis of HAT. This will increase capacity of involved investigators and health personnel.

gHAT is currently a minor public health problem in Africa, but the STROGHAT experience can be useful for tackling other diseases.

Hence the consortium partners will assure that the STROGHAT experience is largely shared among the African research community. ITM and IRD have an extended research network in Sub-Sahara Africa and conduct a post-graduate teaching program on disease control where the STROGHAT experience will be used as example. The HAT platform, chaired by DNDi, provides opportunities to present and discuss experiences among African researchers in gHAT. INRB and PNLTHA staff are involved in teaching and research at universities in Kinshasa.

2.3 (Four year) follow-up of STROGHAT

All partners commit to maximize efforts to ensure that resulting health technologies and services will be broadly available and accessible, which probably will have to be longer than the four-year follow-up as requested by EDCTP.

In principle, both BMGF and the Belgian Government have already committed to support gHAT control in DRC until 2030. Both donors were involved during the preparation of STROGHAT, through in-kind contribution or indirectly by enabling STROGHAT to use existing equipment (see Table 3.1h/PNLTHA and Table 3.1j). Both donors will also closely follow-up the results, which will be used to prepare the new phases of their gHAT interventions. The follow-up to ensure that results will be integrated at policy level have been explained above.

All gHAT drugs are currently donated by Sanofi and Bayern free of charge via WHO. It is the intention that Sanofi will also provide acoziborole free of charge, once EMA has approved acoziborole. Such donations ensure that screening and treatment of gHAT can be maintained at no cost for the population at risk.

2.3 Summary : Key Elements of the Impact Section

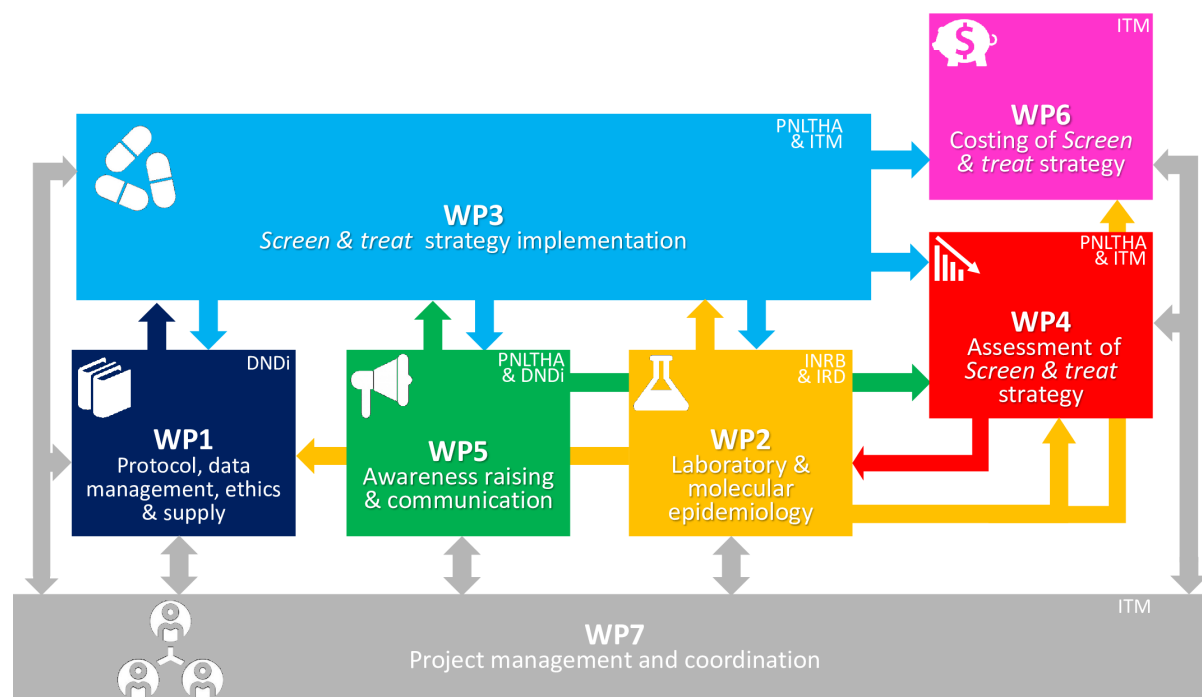
SPECIFIC NEEDS - Current approach of gHAT control not adapted to new epidemiology and non-sustainable, with risk of resurgence - Need for evidence of efficacy of extended acoziborole application before recommendation by WHO - More safety data needed to justify more extended use of acoziborole - Solid costing data needed to develop a long term funding plan required to reach elimination	EXPECTED RESULTS - Evidence of effectiveness of the <i>Screen & treat</i> approach - Additional information on safety of treatment with acoziborole - Improved insight in cost of implementing a <i>Screen & treat</i> approach	D & E & C MEASURES - Exploitation: Costing tool will be available to control programs of gHAT endemic countries and their funders - Dissemination towards the scientific community and health authorities of HAT endemic countries: Scientific publications with the results of the trial in terms of effectiveness and cost of a <i>Screen & treat</i> approach. Participation in key international gHAT control and NTD fora. - Communication towards population of HAT affected areas: Feedback of results of social study
TARGET GROUPS - The results of the project will be up-taken by WHO, health authorities of gHAT affected countries and the donor society - Beneficiaries will be the population of all areas where gHAT has been endemic.	OUTCOMES - Obtained data will contribute to acceptance and recommendation by WHO of wider use of acoziborole - <i>Screen & treat</i> strategy based on acoziborole is adopted as the new state-of-the-art gHAT elimination strategy - Long-term funding plan to progress to elimination of disease in all affected countries	IMPACTS The wider impact is linked to reaching the general objective of interruption of gHAT transmission, at different levels - Scientific: incentive to further invest in NTDs - Economic: Increased productivity in rural sub-Saharan African regions by removing a disease threat that limited economic activities (e.g. fishery and agriculture). - Societal: good news story will boost the health system.

3. Quality and efficiency of the implementation

3.1 Work plan and resources

STROGHAT will be implemented during a period of 5 years between 2023-2027 and is grouped into seven **work packages** (WP) covering all aspects of the project, including management and coordination. All work packages will contribute to the primary and secondary objectives of STROGHAT. An overview of the WPs is presented in Tables 3.1a-j.

Pert chart of the STROGHAT project



WP1 “Protocol development, Data management & Ethics of the clinical study” will finalize the study protocol of the clinical study and submit it for ethics approval in DRC. WP1 will ensure an appropriate monitoring and data management system, and includes training of staff in GCP, study procedures and SOPs. DNDi will ensure availability of acoziborole.

WP2 “Laboratory & molecular epidemiology” will ensure the capacity to implement all laboratory testing foreseen in the project, including the organization and support of specimen management. Also a quality assurance system will be set-up.

WP3 “Screen & treat implementation” will finalize the preparations by selecting the health facilities involved in the study, mapping of all villages of the study area, training of staff and operational planning of the mobile teams. During 3 consecutive years, mobile teams will screen all villages to include seropositives that are eligible in the study and treat them with acoziborole. Seropositives detected in selected health facilities will also be offered to enter the study. All sero-positives not included in the clinical study will be managed according to the national guidelines of gHAT in DRC.

WP4 “assessment of Screen & treat strategy” will assess the impact of the implementation of the *Screen & treat* strategy in year 4. It will screen all villages that have been visited in WP3 during the previous 3 years. An additional sample of 150,000 individuals will be screened in the wider study area.

WP5 “Awareness raising & communication” will ensure maximal participation of the population in the study. The acceptability of the population will be assessed, an appropriate strategy will be developed

and implemented. Yearly the participation of the population will be evaluated and measures will be taken if necessary.

WP6 “Costing of *Screen & treat* strategy” will provide an overview of the costing of the implementation of a *Screen & treat* approach, in view of an extension of the implementation in DRC and other gHAT affected countries.

WP7 will ensure project management, coordination and dissemination of results.

Capacity development at individual, organizational and institutional level is an integrated component of all work packages, focusing on the 2 Congolese institutions (PNLTHA and INRB) and the local research capacity of DNDi in DRC.

Table 3.1a: List of work packages

Work package No	Work Package Title	Lead Participant No	Lead Participant Short Name	Person-Months	Start Month	End month
1	Protocol development, data management, ethics and acoziborole supply	5	DNDi	117	1	42
2	Laboratory & molecular epidemiology	3 & 4	INRB & IRD	233	3	56
3	<i>Screen & treat</i> strategy implementation	1 & 2	PNLTHA	146	1	42
4	Assessment of <i>Screen & treat</i> strategy	1 & 2	PNLTHA & ITM	56	40	60
5	Awareness raising & communication	2 & 5	PNLTHA & DNDi	62	1	56
6	Costing of <i>Screen & treat</i> strategy	1	ITM	5	12	46
7	Project management and coordination	1	ITM	82	1	60

GANTT chart of the STROGHAT project

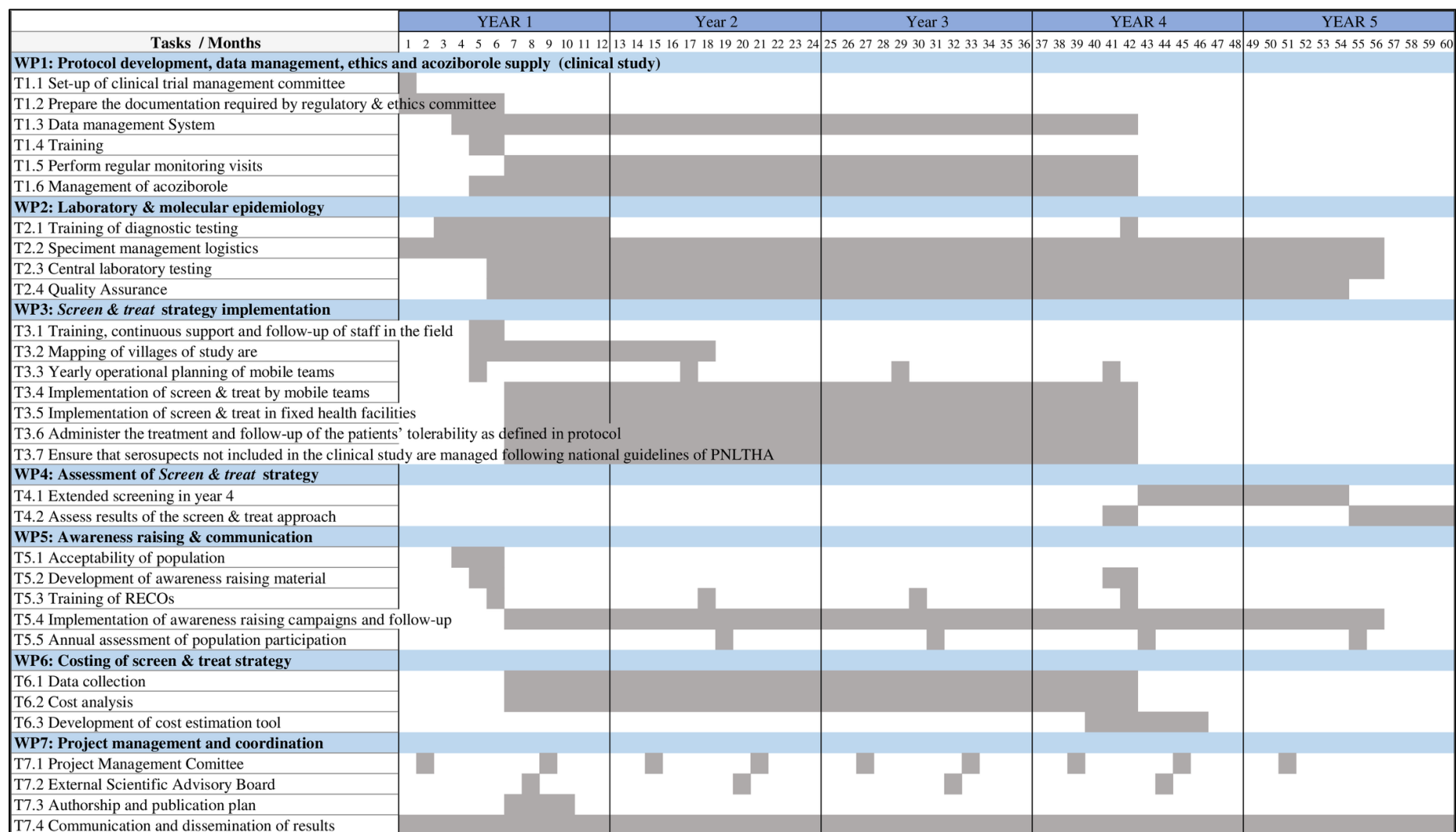


Table 3.1b: Work package description

Work package number	1 (DNDi)
Work package title	Protocol development, data management, ethics and acoziborole supply (clinical study)

Objectives

1. Obtain ethical approval of the clinical study of STROGHAT
2. Ensure compliance with protocol, Standard operating procedure (SOPs) and Good Clinical Practices (GCP)
3. Adopt MEDRIO to capture the clinical data, develop the database and ensure data management
4. Ensure regular reports and final clinical study report
5. Ensure availability of acoziborole

Description of work

Task 1.1. Set-up of clinical trial management committee (Month 1)

DNDi will ensure the organization of meetings, ethical and regulatory submission and follow-up of monitoring activities accordance with the ICH-GCP guidelines. All data will be uploaded to a secure server hosted by DNDi, which will also ensure the data assessment.

The responsables of each work package will meet regularly (e.g. once a month) to evaluate the progress of the clinical study. Meeting reports will be forwarded to the Project Management Committee (PMC).

Task 1.2 Prepare the documentation required by regulatory & ethics committee (Month 6)

This task will prepare the documents required for the approval of the clinical study. This includes the: i) the Study protocol; ii) Informed Consent Form (ICF); iii) Investigator brochure and Insurance Certificate, and iv) Investigational Medicinal Product Dossier. These documents will be submitted to the Institutional review board of ITM, the Ethics Committee of the University Hospital in Antwerp (UZA) and the Ethics Committee and Health Authorities from DRC to obtain approval for the study. The study will be registered in clinicaltrials.gov.

Task 1.3. Data management system

For data collection and management of the clinical study, a Research Electronic Data Capture (MEDRIO) will be used as for other phase3 clinical trials implemented by DNDi.

Task 1.4. Training (month 6)

The staff involved in the clinical study will be trained in MEDRIO, in specific Standard Operating Procedures (SOPs) of the clinical study, in STROGHAT protocol requirements, adverse events recording and Good Clinical Practices (GCP).

The team of local DNDi monitors based in Kinshasa will be re-trained on the monitoring to review compliance of the protocol, SOPs and GCP requirements (ITM and DNDi).

Task 1.5 Perform regular monitoring visits

The team of local DNDi monitors based in Kinshasa will perform regular monitoring visits in the field (every 2 months), and guarantee medical supervision, coordination between all clinical trial parties, clinical safety follow-up of all treated participants, and compliance with Good Clinical Practices (GCP). Adverse events will be monitored by the medical monitor and precisions will be obtained via queries if necessary. The DRC local reference regional pharmacovigilance center will be involved in the follow up of adverse events with the aim to test and guarantee sustainable reporting, ready to extend it to other

geographical areas after the end of the study, through nationally established channels.

Task 1.6 Management of acoziborole supply

DNDi will ensure the timely availability of acoziborole, the supervision of the packaging and labelling and shipment of the acoziborole to the DNDi pharmacy in Kinshasa. The pharmacist will ensure the supply to the PNLTHA regional coordination, supervise the storage and supervise distribution to the clinical investigators. This supervision will be documented following the DNDi procedures.

Work package number	2 (INRB &IRD)
Work package title	Laboratory & molecular epidemiology

Objectives

1. Increase capacity in diagnostic testing of field teams and laboratory
2. Organize and support specimen management
3. Implement central laboratory testing
4. Ensure quality assurance of the whole process

Description of work

Task 2.1: Training on diagnostic testing (lead IRD)

Activity 2.1.1: Training of field teams on diagnostic screening (month 3-42):

- Mobile team members: Month 3-12 (3 mobile teams): CATT and/or RDT screening, correct specimen collection and storage, microscopy, image capturing. Intermittent trainings in case internal quality assurance would point to a need. Month 42 (3 mobile teams + 3 mini-mobile teams): parasitological confirmation of HAT by mAECT.
- Health center personnel: Month 3-12: execution of the study RDT, image capturing, microscopy where relevant. Additional trainings in case internal quality assurance would point to a need.

Activity 2.1.2: Training of national reference laboratory staff on diagnostic tests (Month 1-12): Refreshment training at INRB on all laboratory tests carried out at the national reference laboratory during the study (ELISA/*T.b. gambiense*, iELISA, trypanolysis, *Trypanozoon*-RT-qPCR multiplex). Additional trainings in case proficiency testing or external quality control would point to a need, or at request of the laboratory, e.g. after changes of staff.

Task 2.2: Specimen management logistics (lead INRB)

Activity 2.2.1: Specimen collection kit assembly (month 1-48): Assembly of 5000 specimen collection kits at INRB, containing: blood collection tubes, filter paper for preparation of dried blood spots (DBS), and tubes with RNA/DNA shield buffer and barcoded labels. The kits will contain sufficient material to allow additional tests for quality control. Specimen collection kits will be dispatched at least once a year to the field (month 6: 2000; month 18:500; month 30: 500; month 42: 2000).

Activity 2.2.2: Specimen reception (month 1-54): In month 1-6, INRB will be equipped with appropriate hard and software for specimen management. Once study implementation has started as described in WP4 (month 6), specimens from the field will be received on a monthly basis until the end of the field activities (month 54).

Task 2.3: National reference laboratory testing (lead INRB, month 6-56)

The national reference laboratory will test specimens collected in the field. The testing strategy will be step-wise, starting with serological tests followed by molecular tests.

Activity 2.3.1: Serological testing: Parallel screening with iELISA and ELISA/*T.b. gambiense* will be performed on DBS for maximum sensitivity, exploiting the known high specificity of these tests of 98.7-100%. On all iELISA or ELISA/*T.b. gambiense* positives, trypanolysis, the actual reference test for contact with *T.b. gambiense*, and molecular testing for parasite nucleic acids will be performed.

Activity 2.3.2: Molecular testing: On all iELISA or ELISA/*T.b. gambiense* positives, molecular testing on blood or lymph on RNA/DNA shield:

- The *Trypanozoon*-RT-qPCR multiplex: As a DNA target, the minisatellite repeat region (TBR) of *Trypanosoma brucei* is used by employing 2 probes (q177T and q176T). The 18S rRNA detected by the q18S2 probe is used as an RNA target (Van Reet *et al*, in preparation). The human RNaseP gene will be amplified as an extraction control
- For all subjects testing positive in *Trypanozoon*-RT-qPCR multiplex or trypanolysis, a second molecular test 3-4 months after treatment will verify success of parasite clearance by acoziborole. Particular attention will be given to optimal preservation of the RNA by putting in place an adapted cold chain.

Table: Estimation of the number of tests in the national reference laboratory at INRB

Tests	Month 6-18	Month 19-30	Month 31-42	Month 43-54	Total
Seropositives number	1750	500	500	500 + 1500	4750
iELISA	1750	500	500	500 + 1500	4750
ELISA/ <i>T.b. gambiense</i>	1750	500	500	500 + 1500	4750
trypanolysis	53	15	15	60	143
Molecular	53	15	15	60	143
2 nd Molecular	10	3	3	5	21

Task 2.4: Quality assurance

Activity 2.4.1: Standard laboratory operating procedures (laboratory SOPs) (lead IRD; month 1-42): For all field and laboratory tests, paper format laboratory SOPs will be provided before the screening activities begin (month 1-6). Video standard operating procedures will be prepared for CATT, RDT, mAECT and specimen collection (month 1-6), and mAECT and posted on the website.

Activity 2.4.2: Reagent quality control (lead INRB; month 3-48): Each lot of CATT & RDT tests used in the study will undergo a quality control once they arrive in the DRC.

Activity 2.4.3: Internal quality control (lead INRB; month 6-54): CATT and RDT results from the field will be re-read using electronic pictures. Videos of trypanosome presence will be checked as well.

Activity 2.4.4: External quality control (lead IRD & ITM; month 5-54): Before the start of the national reference laboratory analyses for the project (month 5), and during the study (months 12, 24, 36, 48), a proficiency panel of specimens (selection of positives and negatives) will be provided to INRB. The study laboratory analyses will not start unless correct results are obtained. Corrective measures and additional training will be carried out if external quality control results are unsatisfactory.

Among the study specimens sent from seropositive study participants to INRB, 5% (every 20th specimen) will be expedited to IRD and ITM for external quality control. All specimens with positive results in the central laboratory tests will also undergo external quality control under the supervision of IRD and ITM for external quality control (months 12, 18, 24, 30, 36, 42, 48, 54).

Work package number 3 (PNLTHA lead)

Work package title Screen & treat strategy implementation

Objectives

1. Ensure screening coverage of all selected villages during the period of the study
2. Prepare seropositives to participate in the clinical study
3. Ensure safety management during the entire duration of the clinical study (3 years)
4. Ensure that eligible seropositives are treated with acoziborole and followed-up as per GCP
5. Ensure that seropositives that are excluded from the clinical study, are managed according to the national HAT control guidelines

Description of work

Task 3.1 Training, continuous support and follow-up of staff in the field (PNLTHA & ITM)

Training will be provided to PNLTHA staff to ensure the correct implementation of WP3 (correct use of TRYPELIM tool, recording pictures and videos etc.). During the implementation of STROGHAT, there will be continuous support and follow-up of staff conducting activities in the field.

Task 3.2 Mapping of villages of study area (PNLTHA & ITM)

The available population data are based on data from Landcam and WorldPop. During the first study year, all villages in the study area will be mapped and more accurate population estimates will be obtained.

Task 3.3 Yearly operational planning of mobile teams (PNLTHA & ITM)

PNLTHA will develop yearly operational planning of all villages to be visited by a mobile team for HAT screening according to the WHO criteria of list 1 and 2 (see higher). Villages of gHAT cases detected during the period of the study via passive detection will be added to list 1, for the years to come.

Task 3.4 Implementation of *Screen & treat* by mobile teams (active screening) (Months 7 – 54)

During the clinical study (3 years) the mobile teams will screen the population of the selected villages with CATT and offer all seropositives that meet the inclusion criteria, to enter the *Screen & treat* study. The villages will be re-visited daily upto Day 5 and at 3 months after the first visit to ensure follow-up of treated individuals.

A picture will be taken of all positive serological tests, as well as a video of all positive mAECT tests visualizing parasites, to be used for quality control afterwards. Before treatment, a blood specimen will be collected for retrospective analysis. All data will be digitally recorded via the TRYPELIM tool.

Task 3.5. Implementation of *Screen & treat* in fixed health facilities

The 30 WHO sentinel sites that already ensure gHAT surveillance in the region for WHO, potentially expanded by some other health facilities, will also implement the *Screen & treat* strategy during 3 years. The patients visiting these facilities will be selected for screening with RDT (HAT Sero-K-Set) according to WHO guidelines and all seropositives will be proposed to participate in the study. Collected blood specimens will be transferred to the study team.

Task 3.6 Administer the treatment and follow-up of the patients' tolerability as defined in the protocol

All patients will be visited daily, up to 5 days after treatment by an authorized person. From all individuals that tested positive for molecular testing, a blood sample will be taken during the visit after 3 months to test for RNA.

Task 3.7. Ensure that seropositives not included in the clinical study are managed following national guidelines of PNLTHA

Seropositives of both active screening and passive screening that do not agree to participate in the clinical study or that do not meet the inclusion criteria will be managed according to the national gHAT control strategy of DRC.

Work package number	4 (ITM & PNLTHA lead)
Work package title	Assessment of <i>Screen & treat</i> strategy

Objectives

1. Assess the impact of the *Screen & treat* strategy after 3 years

Description of work

Task 4.1 Extended screening in year 4

In the 4th screening year, all villages screened by mobile teams in the previous 3 years will be screened again. In addition a cluster sample of 150,000 will be screened in villages where there were no gHAT cases reported before the start of the study. A blood specimen from consenting seropositives will be taken for molecular testing. Seropositives detected in year 4 will be examined microscopically with mAECT for parasite presence and managed further following national guidelines of PNLTHA. Before starting this year, documents will be submitted to the Institutional review board of ITM, the Ethics Committee of the University Hospital in Antwerp (UZA) and the Ethics Committee and Health Authorities from DRC to obtain approval for the epidemiological study.

Task 4.2. Assess results of the *Screen & treat* approach

The assessment will focus on comparison of prevalence before and after the study in the 63 villages, and whether or not there is indication of interruption of transmission of *T.b. gambiense* based on data from a cluster random sample.

Work package number	5 (PNLTHA & DNDi lead)
Work package title	Awareness raising & communication

Objectives

1. Maximize participation of population to the study in selected area
2. Ensure awareness raising of population in selected area
3. Ensure participation of local authorities and health providers

Description of work

Task 5.1 Assess acceptability of population

At the start of STROGHAT, a study will be conducted by social scientists from PNLTHA and ITM to explore the acceptability of the STROGHAT procedures and factors associated with participation by the population. Qualitative data will be collected through focus group discussions, interviews, and observation. Gender, age, social groups, and village levels of participation will be considered and thematic areas such as perception of the door-to-door screening, barriers, and difficulties faced for participation, as well as enabling factors explored. Stigmatization, perception and understanding of the HAT control tools applied will be dealt with. These findings will be used to advise on an optimal strategy to achieve maximum participation in the trial.

Task 5.2 Develop appropriate awareness raising material

Tools will be developed to raise community awareness and enhance participation, taking into account results of the acceptability study. Already available materials will be reviewed, feedback obtained from the communities, and material will be adapted according to identified needs, and the new diagnostic strategy and treatment. All tools developed will be re-presented to the communities to obtain their feedback/approval.

Task 5.3 Implementation of RECOs and local associations and organizations

Training on materials will be organized at provincial and then health zone including opinion leaders *Relais Communautaires* - RECOs and other local associations and organizations. All village health workers of the selected villages will be trained about the new *Screen & treat* approach and the adapted awareness-raising tool. Attention will be given to the consequences of the innovation (e.g., no lumbar puncture, treatment on the spot, etc.). Halfway the study, a refreshing course will be organized.

Task 5.4 Implement awareness raising campaigns and conduct follow-up

The RECOs in charge of the selected villages will visit the villages shortly before the arrival of the mobile teams and conduct awareness raising sessions to encourage the population to participate in the study. The RECOs will also facilitate the follow-up of treated individuals.

Task 5.5 Annual assessment of population participation

Based on results obtained from the TRYPELIM tool, the population participation will be yearly assessed and action taken if needed. Any adaptation of the material, training, information and strategy will be reported.

Work package number	6 (ITM lead)
Work package title	Costing of <i>Screen & treat</i> strategy

Objectives

1. Obtain insight of costing of *Screen & treat* strategy in DRC
2. Provide data for mathematical modeling
3. Provide costing data of a *Screen & treat* strategy for HAT control programs outside DRC

Description of work**Task 6.1. Data collection**

WP6 will collect data regarding the cost to implement the *Screen & treat* strategy in the study area from the healthcare provider's perspective. A mixed costing methodology will be used to estimate both the financial and economic costs. Bottom-up micro-costing of activities will be used to evaluate the direct costs at the implementation level. Each resource will be identified and valued according to the purchase price or replacement value. Top-down gross costing will be used to estimate costs that cannot be directly allocated to the field activities, such as training and management costs.

The study will adopt a healthcare provider's perspective focusing on costs incurred by the Ministry of Health (PNLTHA and INRB) and donors. Only costs related to the STROGHAT elimination strategy will be considered and research and management costs will be excluded.

Task 6.2 Cost analysis

The overall financial and economic cost of the intervention will be estimated. The cost per person covered will be estimated by dividing the overall cost by the study population, including the screening via mobile teams and via health centers. The impact of changes in the principal cost divers will be assessed and uncertainty will be factored in to provide strategic recommendations for cluster locations.

Task 6.3 Development of cost estimation tool

The cost information collected will be used to develop a tool to help national programs estimate the cost and budget necessary if they would like to implement a similar intervention in a different context if the trial shows promising results.

Work package number	7 (ITM lead)
Work package title	Project management and coordination

Objectives

1. Coordinate and monitor all WPs, tasks, and activities
2. Provide administrative, technical, and scientific support to the consortium
3. Ensure communication with key counterparts and stakeholders
4. Ensure optimal management and dissemination of project results

Description of work**Task 7.1. Project Management Committee**

For the day-to-day management, a Project Management Committee (PMC) will be set-up involving the five partners of STROGHAT. Under the lead of Epco Hasker, the PMC will be responsible for making key decisions regarding content, finances, intellectual property, consortium changes and appointments. This committee will monitor all aspects of project progress, approve annual work plans from partners, and approve the annual distribution of funds. A consortium agreement will be signed. They will meet regularly (e.g. quarterly) to discuss the progress or *ad hoc* when needed. The meetings will be organized by ITM and meeting reports will be developed and distributed.

ITM will ensure that each work package (WP) is implemented, monitored, and managed according to pre-determined plans and standards. Each WP leader will be responsible for the implementation of his or her respective WP and will make short-term, lower-level decisions that do not require consensus among all partners. He or she will need the approval of all consortium partners on any key decisions, such as those directly related to deliverables and milestones.

Throughout the STROGHAT project, regular virtual meetings and in-person team meetings will be organized, i.e. at least biannually. At the start of the project, we foresee a face-to-face kick meeting to which all partners, as well as members of the scientific advisory board (see below), will be invited. During the project, virtual meetings will be organized every quarter, in addition to two face-to-face meetings.

Task 7.2. External Scientific Advisory Board

In addition to the Project Management Committee (PMC), an external *Scientific Advisory Board* will be established. Key experts in the field of HAT and NTD control will be invited to become members. The HAT experts of the WHO department of control of neglected tropical diseases have already agreed to participate. Meetings will be conducted at least once a year to discuss results.

Task 7.3. Authorship and publication plan, data management plan and stewardship and global access plan

In the first year of the project an authorship and publication policy and plan and a data management plan will be developed. A stewardship and global access plan will be developed at months 30 and 60.

Task 7.4. Communication and dissemination of results

The large public will be informed about the progress and results of STROGHAT through a website managed by ITM, including a Twitter account. Press releases will be sent out and short articles will be published through different media whenever relevant. Abstracts concerning the study will be submitted to major conferences. Outcomes of the study will be disclosed in meeting presentations, and in the form of publications in high-tier open-access journals. At the end of the project, a dissemination meeting will be held in which project results will be presented to a wide variety of stakeholders.

Table 3.1c: List of deliverables

Number	Deliverable name	Short description	WP	Lead	Type	Dissemination level	Delivery month
D1.1	Study initiation package	Protocol of clinical study registered and approved by ethics committee in DRC	1	DNDi	Ethics	PU	6
D1.2	Training report	Training report on data management, GCP and SOPs	1	DNDi	R	PU	6
D1.3	Monitoring	Yearly monitoring report of clinical study	1	DNDi	R	SEN	20, 32, 44
D1.4	Midterm recruitment report	Overview of recruitment by mobile team, problems in recruitment and measures to compensate	1	DNDi	R	PU	24
D2.1	Laboratory	Laboratory SOPs for field and laboratory	2	IRD	DEC	PU	6

	SOPs	testing available on paper and/or video from website					
D2.2	Training report	Diagnostic training provided to field and national reference laboratory staff.	2	IRD	R	PU	12, 42
D2.3	Test Reports	Reports on specimens received and tests performed at INRB and reports on external quality control results	2	INRB	R	PU	18, 30, 42, 56
D2.4	Internal quality assessment	Report on reading of pictures and videos	2	INRB	R	SEN	18, 30, 42, 54
D3.1	Training report	Training report for treatment and follow-up	3	PNL THA	R	PU	6, 24
D3.2	Operational planning	Yearly operational planning documents	3	PNL THA	R	PU	5, 17, 29, 41
D3.3	Active Screening	Annual reports of screening by mobile teams (month)	3	PNL THA	R	PU	20, 32, 44
D3.4	Passive screening	Annual reports of passive screening in selected health facilities	3	PNL THA	R	PU	20, 32, 44, 56
D3.5	Midterm recruitment report	Number of participants by clinical sites, problems report, and measures to compensate for delays	3	PNL THA	R	PU	24
D4.1	Screening report	Report of screening in year 4	4	PNL THA	R	PU	56
D5.1.	Acceptability	Report on the acceptability assessment	5	ITM	R	PU	6
D5.2	Strategy	Report of the community-designed awareness strategy	5	DNDi	R	PU	6
D5.3	Health Promotion tool	New Health Promotion tools developed	5	DNDi	Other	PU	12
D5.4	Population participation	Yearly assessment reports of population participation and adaptation	5	DNDi	R	PU	20, 32, 44
D6.1	Costing Protocol	Approved costing study protocol by ethical committee	6	ITM	Ethics	PU	12
D6.2	Costing report	Report on financial and economic cost of the intervention	6	ITM	R	PU	44
D6.3	Costing tool	Cost estimation tool for used in other settings	6	ITM	Other	PU	48
D7.1	Publication	Authorship and publication plan	7	ITM	DEC	PU	12
D7.2	DMP	Plan detailing all aspects of data management	7	ITM	DMP	PU	12
D7.3	Stewardship plan	Plan how to achieve optimal use of STROGHAT	7	ITM	DEC	PU	30, 60
D7.4	Global access plan	Appropriate and proportionate plan	7	ITM	DEC	PU	60
D7.5	Dissemination	End meeting report	7	ITM	R	PU	60
D7.6	Result posting status	Results posting in the applicable registry/ies	7	ITM	R	PU	60

Table 3.1d: List of milestones

Milestone number	Milestone name	Related WP	Due date (in month)	Means of verification
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M1.1	Study protocol ready for ethical and regulatory submission	1	3	Document available
M1.2	Data capture and database ready for inclusion of study participants	1	4	Up and running
M1.3	Annual investigator meeting and training	1	4	Meeting report available
M1.4	1 st patient treated with acoziborole	1	6	Report available
M2.1	Specimen collection kits expedited from INRB to field	2	6, 18, 42	Report available
M2.2	Reagent quality checked	2	6, 18, 30, 42	Report available
M3.1	Map of updated population estimation of the villages of the study area	3	18	Document available
M5.1	Existing HAT-related health promotion material compiled	5	3	Report available
M7.1	Kick-off meeting	6	2	Report available

Table 3.1e: List of critical risks.

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	Likelihood	Severity	WPs involved	Proposed risk-mitigation measures
Political instability/civil unrest in DRC impacting implementation of project activities.	M	M	1, 2, 3, 4, 5	Security situation was a criterion for the selection of study area. Close ties will be maintained with staff & authorities to monitor the security situation. Establish distance-based assistance, if required.
Emerging or reemerging infectious diseases interrupt the implementation of the project.	M	M	1, 2, 3, 4, 5	The public health situation is monitored closely by Congolese partners. Establish distance-based assistance, if required.
Delay in EC & regulatory approval process	M	M	1, 2, 3, 6	Remain in close contact with local authorities.
Timely availability of acoziborole	L	H	3,4	The acoziborole supply for STROGHAT will be obtained from excess production for other clinical trial – this production process has been started and is followed up by DNDi BMGF will contribute in kind to cover additional costs (e.g. packaging) Sanofi offered to assist DNDi if needed (e-mail available)
Commercial companies stop production of rapid diagnostic tests	M	L	3, 4	Replace RDTs by CATT which is produced by ITM
Fuel shortages in study area owing to Russia-Ukraine war.	M	M	1, 3, 4, 5	Proactive planning of supply needs.
Delays in shipment & importation of supplies (e.g. reagents, tests, drugs, equipment, etc.) required for project activities.	M	M	1, 2, 3, 4	Proactive planning of supply needs & shipments. Ensure sufficient & adapted storage capacity. Ensure timely submission of import applications and maintain close contact with customs authorities.
Participation in screening	L	H	3,4	Awareness-raising & communication

insufficient to achieve desired outcome.				tools promote key messages and enhance participation of the population from selected areas.
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L = Low, M = Medium, H = High

Table 3.1f: Table showing number of person months required

Person / Months	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total Person-Months per Participant
1. ITM		8	20	8	2	5	34	77
2. PNLTHA			126	48	12		12	198
3. INRB		216					31	247
4. IRD		9					3	12
5. DNDi	117				48		2	167
Total	117	233	146	56	62	5	82	701

Table 3.1g: ‘Subcontracting costs’ items

5/DNDi (associated partner) these costs will be financed by Switzerland		
Subcontracting	Cost (€)	Description of tasks and justification
UBC	16,000	Pharmaco vigilance support

Table 3.1h: ‘Purchase costs’ items (travel and subsistence, equipment and other goods, works and services)

Participant 1/ITM	Cost (€)	Justification
Travel and subsistence	241,600	Short travel 4x3,700€/yr (5yrs) Long stay in DRC 4x8,380€/yr/5yrs
Equipment	84,330	Vehicle for supervision and follow-up 50,000€ / replacement of equipment of mobile teams 34,330€
Other goods, works and services	671,600	Meetings 30,000€ / Serological tests (CATT, ELISA, iELISA, Trypanolyse) 460,600€ / mAECT 25,000 € / molecular tests 51,000€ / Shipment 90,000€/audit 15,000
Total	997,530	

Participant 2/PNLTHA		
	Cost (€)	Justification
Travel and subsistence	114,000	WP3 &4 Long travel to study site (2x2 per year) 4,800€x20 WP7 Participation meetings in Europe 3,000€x6
Equipment		Equipment for mobile teams (including vehicles) have been budgeted in ongoing HAT support project supported by Belgian Government (2022-2026)
Other goods, works and services	514,200	WP3&4: operational planning workshops 8,000€x4 WP3&4 Shipment costs 15,400€x4 years; Operating costs 74,400€x4years; Fuel 20,250€x4 years; other 20,000€ WP7 International meeting organization in DRC 25,000€x2
Total	628,200	

Participant 3/INRB		
	Cost (€)	Justification

Travel and subsistence	93,500	WP2 Long travel to study site (30 days) 7,000€x8; WP2 Long travel to Europe (30 days) 7,500€x3; WP7 Participation meetings in Europe 3,000€x6
Equipment	30,500	WP2 ELISA washer & reader 2,000€; WP2 Centrifuge 5,000€; WP2 Freezer -20°C 1,500€; WP2 Laptop associated to lab equipment (ELISA & qPCR) 1,000€x2; WP2 Hard & software for specimen management 4000€;
Other goods, works and services	88,000	WP2: Lab consumables (plastic ware & reagents) 42,000€; Material training mobile teams & health center staff 15,000€; Shipment costs (Kinshasa-Europe) 12,000€; audit 8,000€; Shipment costs (Kinshasa-Equateur) 3,000€; Communication costs 6000€; Tube labels -70°C 2000€;
Total	212,000	

Participant 4/IRD		
	Cost (€)	Justification
Travel and subsistence	53,000	WP2 3x long travel to study site via Kinshasa (1 month) 8,500€x5; WP2 5x short travel to Kinshasa (10-days) 3,500€x5; WP7 Participation in 2 African meetings 3,000€x2; WP7 Participation in 2 European meetings: 2,000€x2;
Other goods, works and services	38,500	WP2: Lab consumables (plastic ware, reagents) 12,500€; External quality control trypanolysis 8,000€; External quality control molecular testing 8,000€; EQA panel transport 6,000€; External quality control ELISA screening 4,000€;
Total	91,500	

5. DNDi (associated partner) This budget will be provided by the Gov of Switzerland		
	Cost (€)	Justification
Travel and subsistence	104,729	Budget for local travel for CT Manager, Pharmacist and 4 Clinical Research Associates during the implementation of the project (€53K), key Partners Meeting travel and logistics costs - one per year (€25K), and One Investigator Meeting travel & logistics costs (€27K)
Other goods, works and services	271,544	Database MEDRIO costs (€96K), development of the training protocol and management of adverse events (€60K), technical assistance services for the preparation of specific study documentation (€45K), networking campaign to build awareness in local communities (€21K), local support and tools for the proper implementation of activities (e.g. computers, printers, adaptation and printing of training material, ..)(€15K), clinical trial insurance (€20K), and mandatory audit fees (€15K)
Total	392,273	

Table 3.1i: 'Other costs categories' items (e.g. internally invoiced goods and services)

1/ ITM		
Internally invoiced goods	Cost (€)	Justification
CATT	370,500	Purchase of 650,000 tests manufactured at ITM

Table 3.1j: Table showing in kind contributions

Participant Number/Short Name			
Third party name	Category	Cost (€)	Justification
BMGF	Other goods, works and services	50,000	BMG will contribute to manufacturing by ensuring clinical packaging, labelling, and shipment.

3.2 Capacity of participants and consortium as a whole

The STROGHAT consortium is composed of five organizations which have both the scientific and management capacity to ensure all the tasks required to implement STROGHAT and ensure that results will be used and implemented in the future.

In August 2022, the candidacy of DRC to become EU associated state has been accepted. PNLTHA and INRB are the 2 main key-players of DRC in gHAT control. They managed the activities that led to active decline in the number of cases in recent years. PNLTHA has a proven capacity to screen millions of individuals per year, even in hard conditions, and it has a long tradition to conduct control activities in the selected area. PNLTHA has been involved in numerous operational research including with all other partners of the consortium. As it has the mandate to recommend new gHAT control strategies to the Ministry of Health, it will play a major role in the follow-up of the results of STROGHAT.

The INRB is the national reference laboratory for gHAT in DRC, and is recognized by WHO as a collaborating center for gHAT diagnostics (the only one for gHAT in Africa). The parasitology department has closely collaborated with ITM, IRD and DNDi for over 20 years in the framework of gHAT research, diagnostic development and drug development. INRB is the national reference laboratory for HAT but will also play an increasing role in the routine HAT control program in the following years, in particular for training of laboratory staff, confirmatory testing, quality assurance of HAT diagnosis and operational research. It has the capacity to perform all available serological and molecular tests. Its human resources will be increased to enable to perform the expected additional workload described above within a limited time frame. INRB has the capacity to prepare all individual kits that will be used for sampling in the field.

The 3 European institutions are major international key players of the international HAT community, and have played a major role in the recent progress of gHAT control. ITM coordinates the DRC control program for the Belgian Government and the *Bill & Melinda Gates Foundation* (BMGF), the major source of funding in support of the national HAT control program of DRC. IRD actively supports the national control programs of Côte d'Ivoire, Guinea, Burkina Faso, Cameroun and Chad, for which it is also supported through the BMGF. Its involvement with other countries control programs, will also facilitate uptake of STROGHAT results. IRD and ITM have a longstanding collaboration and reputation in developing, evaluation and even production of HAT diagnostics. Also IRD and ITM are recognized collaborating centers of WHO for gHAT diagnostics, which recognizes fruitful collaboration with WHO over the previous years.

DNDi is based in Switzerland and has been asked to join the consortium as associated partner. The reason is that DNDi is the only organization that can implement the clinical study. DNDi has participated in all new treatment trials (NECT, Fexinidazole and acoziborole) since 2009. Dndi was involved in important capacity building and training since 2009. The research department of DNDi was able to bring to the clinical staff the first new oral treatment fexinidazole actually included in the WHO gHAT treatment guidelines as the first treatment of gHAT patients. In STROGHAT, DNDi is in charge of ensuring the timely availability of the drug, a responsibility only DNDi can ensure. As DNDi cannot be beneficiary of Horizon Europe grant, Government of Switzerland has committed to finance the budget of DNDi required to obtain the expected results.

ITM and DNDI have a registered office in Kinshasa, DRC, exempted from import taxes and are used to deal with logistics including import and transport of equipment and consumables.

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STOP Transmission of gambiense Human African Trypanosomiasis (STROGHAT)

INFORMATION ON CLINICAL STUDY

1 Description

Stop transmission of gambiense human African trypanosomiasis' (STROGHAT)

1.1 Study rationale

Human African trypanosomiasis (HAT) is a vector borne parasitic disease, fatal if left untreated, affecting rural communities in sub-Saharan Africa.[1] The parasite is transmitted by tsetse flies of the genus *Glossina*. [2] Since the 19th century, HAT is known to have caused several devastating epidemics. It is estimated that at the turn of the 20th century 300,000 to 500,000 people died of HAT, with another epidemic occurring in the 1920s-‘30s.[3] By 1960 intensive efforts had brought the disease under control again in most countries but when measures were neglected it reemerged. Hundreds of thousands have perished from HAT as recently as the 1990s.[4] Currently incidence is at an all-time low but it has very low before. If left unattended, HAT has a strong tendency of resurgence. [5] Thanks to recent innovations we may now be in a position to achieve a more permanent state of control and even eliminate transmission of *Trypanosoma brucei gambiense*, the parasite causing over 90% of HAT cases. We propose a trial in which we will pilot a ‘screen and treat’ strategy for gambiense human African trypanosomiasis (gHAT), making use of a new single dose oral treatment. Our aims are to demonstrate that the new drug, acoziborole, can safely be used in serological gHAT suspects and that by systematically doing so we can arrive at zero prevalence, i.e. elimination of transmission, in the area targeted.

Until one decade ago, treatment options for gHAT were very limited and very toxic, forcing control programs to avoid overtreatment at all costs.[6] An estimated five to ten percent of patients treated for second stage HAT died from the treatment, which was based on an arsenic derivative. [7] This necessitated complex diagnostic procedures including screening with a serological test followed by laborious microscopic confirmation of seropositives on the spot and determination of disease stage which required a lumbar puncture. In each step of the diagnostic chain, considerable numbers of potential HAT cases (an estimated 50% altogether) were lost and therefore remained untreated.[8] Because humans are the reservoir for gHAT, not treating substantial numbers of infected persons will lead to sustained transmission and reemergence, as observed in the past.

Recently a non-toxic single dose oral drug, acoziborole, has passed phase 3 evaluation.[9] This drug can treat gHAT irrespective of disease stage, thus rendering the lumbar puncture for stage determination redundant. Having a single dose oral drug available for both stages of the disease opens up new perspectives for eliminating gHAT. Treating anyone testing positive to a serological screening test, without further diagnostic confirmation and stage determination, will avoid missing many cases and will reduce the complex procedures needed to confirm the infection.

Acoziborole has been developed by Drugs for Neglected Diseases initiative (DNDi) in collaboration with Sanofi. Preliminary results of a clinical trial suggest adequate safety and efficacy in the treatment of confirmed gHAT cases. A potential wider use of this drug, provided safety and efficacy are confirmed, has been discussed in technical gHAT groups since several years. This includes the possibility of treating serological suspects without parasitological confirmation. At a WHO meeting on “Integration of new tools into national and global policies” (Dec 2019) a Target Safety Profile was agreed upon, presented in Table 1.

Table 1: Target Safety Profile for escalating extension of use of a single-dose oral medicine

Scenarios	Advantages/opportunities	Disadvantages/challenges	TPP safety
<i>1. Treatment of confirmed gHAT cases</i>	- Clear benefit for people treated - Known number of true cases - Good control of drug use - Limited drug pressure averts drug resistance	- Requires parasite exam for confirmation (hinders integration) - A certain number of infected (but unconfirmed) people are not identified and remains as reservoir	<i>Comparable to current treatments (pentamidine, NECT, fexinidazole)</i>
<i>2. Treatment of confirmed gHAT cases + suspects</i>	- Coverage of confirmed cases and other human parasite carriers - Reduces transmission better/faster than scenario 1 - Good control of drug use - Limited drug pressure averts drug resistance - Known number of true cases, hence it's easier to know when to stop such extended use	- Still requires exam to study suspects in order to confirm cases (hinders integration) - Some people receive unneeded medicine (depending on the PPV of the diagnostic method used)	- More robust safety data, n larger cohorts (e.g. from pharmacovigilance) - Safety data in pregnancy - Safety data in children (ideally all ages, but at least in >5 years)
<i>3. Treatment of confirmed gHAT suspects without parasitological exams</i>	- Coverage of many human parasite carriers including cases depending on diagnostic method used - May reduce transmission better/faster than scenario 1 - No need of parasitology work - Facilitates integration in the health system - Good control of drug use	- Some people receive unneeded drugs (depending on the diagnostic PPV) - Benefit/risk ratio for individuals treated, lower than scenario 1, and with time lower than scenario 2 - Uncertainty about HAT epidemiological situation (data on confirmed cases) hence difficult to know when to stop this approach	<i>Strong Safety data</i> - Larger studies detecting uncommon adverse events (AE) (1/100 to 1/1,000) - Safety data in pregnancy - Safe in children (ideally all ages, but at least in >5 years)
<i>4. Collective use in well-defined populations</i>	- Broader coverage of human parasite carriers including cases - Reduces transmission even faster than scenario 1, 2 and 3 - No need of screening or confirmation tests for the intervention itself - Facilitates integration with other community-based health activities	- Most people receive unneeded medicine - Benefit/risk ratio for individuals treated, lower than scenario 1, 2 and 3 - Possible risk of uncontrolled use of the drug - Requires population acceptability - Involves complex outreach work - Poor epidemiological visibility	<i>Requires larger studies and specific studies to complete safety knowledge</i> - Rare AE (1/1,000 to 1/10,000) studied and characterized - Good safety in pregnancy - Safe in children (ideally all ages but at least in >5 years old) - No interactions with other drugs or pathogens, causing toxicity or threatening efficacy of other drugs

5. Mass administration in endemic areas	- Treatment of all parasite carriers - Should strongly reduce transmission - No need of screening or confirmation tests for the intervention itself - Facilitates integration with other community-based health activities	- Most people receive unneeded medicine - Benefit/risk ratio is very low - High drug pressure. Risk of uncontrolled drug use and resistance - Requires population acceptability - Involves complex outreach work - No epidemiological visibility	- Proven very low toxicity - no attributable SAE - Very rare adverse events (<1/10,000) studied and characterized - No toxicity in children exposed in utero - No threat in pregnancy - Safe in children and elderly - no drug interactions causing toxicity or threatening efficacy of other drugs
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This widened use implies treating a certain number of individuals not infected; hence it would require evidence of a good benefit-risk ratio. In order to build the evidence required, larger clinical trials are needed to generate safety data in gHAT suspects unconfirmed by parasitology. The studies need to be sufficiently powered to cover most of the safety spectrum, including uncommon adverse events (AE). During the tenth meeting of the sub-group dedicated to explore the possible uses of a single dose oral drug in the context of elimination of gHAT, it was agreed that DNDi would present to WHO a study designed to generate the evidence needed. Further to this meeting, a new phase II/III study double blind versus placebo study in seropositives started in December 2021 in 5 sites in DRC and 1 site in Guinea. Thousand two hundred seropositive suspects will be allocated in an unbalanced randomization to Acoziborole administration (900) or placebo (300). The study is planned to be completed in Q1, 2023.

The STROGHAT study will provide additional safety data and will also allow to document cost of the screen & treat approach, which will be of key importance for a decision to roll it out on a larger scale. This is conform scenario 3 of Table 1, i.e. *Treatment of confirmed gHAT suspects without parasitological exams*.

Although this innovative option of ‘screen & treat’ to eliminate gHAT transmission is now within reach, its true feasibility and effectiveness under field conditions remain to be confirmed and optimal implementation modalities need to be determined. We hypothesize that by systematically screening the populations of all endemic villages in high risk areas and treating with acoziborole all serological suspects identified over a three-year period, we will be able to arrive at a zero prevalence across the entire area. Apart from demonstrating that a ‘test & treat’ strategy with acoziborole can eliminate gHAT transmission, the STROGHAT study will also provide complementary safety data in serological gHAT suspects under real field conditions.

For this project, an area has been selected which according to the WHO definition is still at moderate to high risk for gHAT, i.e. an estimated annual risk of infection of 1/10,000 or above and which is surrounded by areas of low or negligible risk. Within the area there are 63 villages that would need to be screened by mobile teams according to the current strategy, 48 because they had incident cases in the last three years (2018-2020, list 1 of WHO) plus 15 that had no cases in the last three years but did have cases less than five years ago (2016-2017, list 2 of WHO). In the selected area, WHO supports 30 fixed health facilities to function as sentinel sites for gHAT surveillance. If a gHAT case is identified, the village of origin of the patient is added to the list of villages to be screened.

We will conduct screening by mobile teams in all the villages of list 1 and 2, which is in accordance with the current strategy. However instead of performing diagnostic confirmation on the spot and treating only parasitologically confirmed cases, we will treat all identified serological suspects with acoziborole, after having collected a venous blood sample and if applicable a gland puncture sample as

well. These samples will be used for post hoc confirmation of infection with *T.b. gambiense* through a molecular test in a remote laboratory. This will be repeated in the second and third project year for villages still eligible (on list 1 and 2). In addition the same screen & treat strategy will also be applied in the sentinel sites and some other selected areas.

During the fourth and final year of the study we will screen all villages that were originally endemic or were added over the course of the study (list 1 and 2), as well as a random sample from villages in the study area that have not been screened by mobile teams but that were covered by the health system via passive screening (total 403 villages, estimated population 2-2.5 million). At this stage we will perform parasitological confirmation on the spot for serological suspects and also draw a venous blood sample (or lymph node aspirate if appropriate) for remote testing. This will allow us to verify whether a zero prevalence has been achieved in the villages targeted for active screening as well as in the wider area.

1.1.1 Extent and evaluation of current knowledge directly linked to the scientific question(s) to be answered by the clinical study

1.1.1.1 Background information:

In its strategic document ‘Ending the neglect to attain the sustainable development goals. A road map for neglected tropical diseases 2021–2030’, WHO has targeted gHAT for elimination of transmission by 2030. [10] A metric to monitor progress towards this target is assessment of areas at moderate to very high risk of infection. [11-14] Such areas are expected to progressively shrink. Very high incidence is defined as an annual infection risk above 1%, high risk is defined as an annual risk between 1% and 1 in 1,000, moderate risk is defined as an annual infection risk between 1 in 1,000 and 1 in 10,000. The size of such areas is being regularly assessed based on cases reported in the preceding 5-year period. The most recent assessment was published in 2022, based on data for the period 2016-2020 (figure 1).[15]

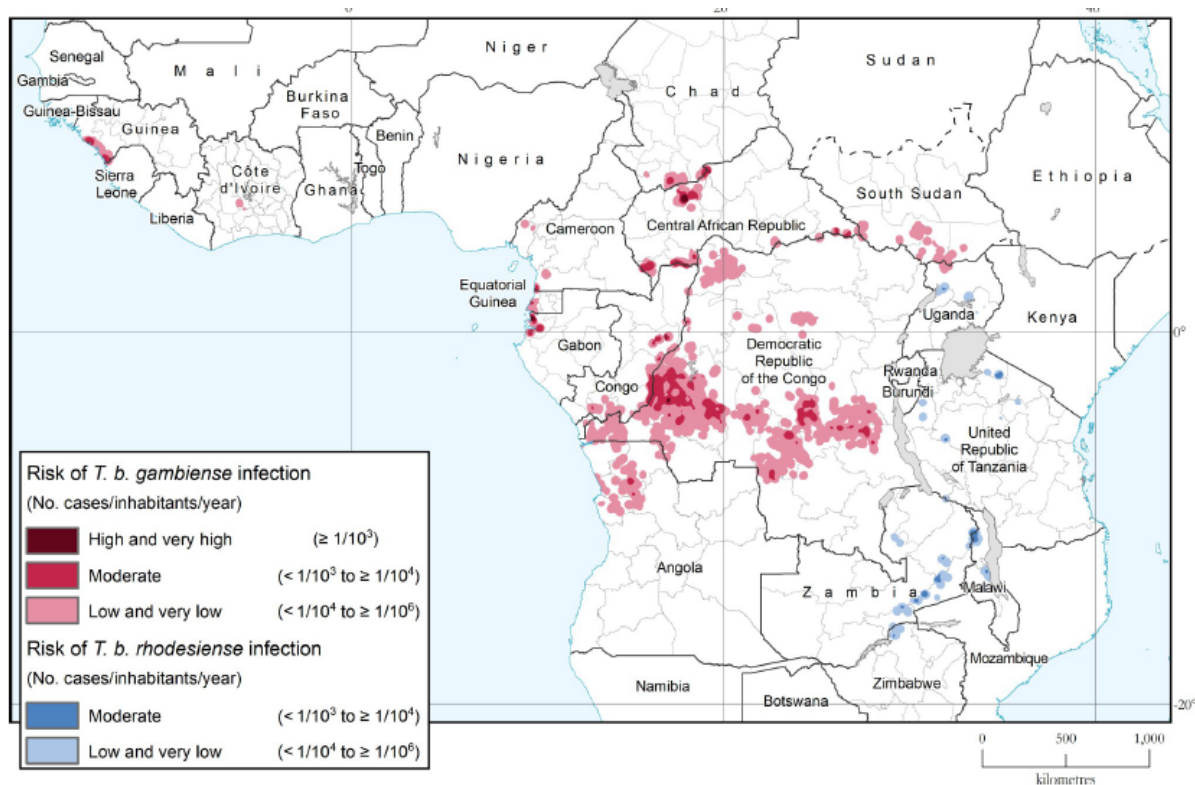


Figure 1: The areas at risk of HAT infection. Period 2016-2020. (Franco et al., PloS NTDs)

As can be seen at one glance from the map in figure 1, the country worst affected is the Democratic Republic of the Congo (DRC).

1.1.1.2 Current gHAT control efforts:

As was already mentioned, HAT is transmitted through the bite of infected tsetse flies. There are two forms but by far the most common is the form of HAT caused by the parasite *Trypanosoma brucei gambiense*, also known as gHAT. [1] The disease has two stages, of which the first stage has few or atypical symptoms. In the second stage the parasite has crossed the blood-brain barrier and more typical and severe symptoms arise, eventually leading to coma and death. Though transmitted by tsetse flies, humans are assumed to be the only significant reservoir of gHAT and therefore interruption of transmission is theoretically possible. A target has been set by WHO for this to be achieved in 2030.[16]

Simarro in 1991 compared different HAT control strategies with and without vector control and concluded that active serological screening followed by parasitological confirmation was the most cost effective approach. [17] Confirmed cases are then referred for treatment. This has since become the main HAT control strategy. However, interrupting transmission will require a different approach from the classical strategy aimed at merely controlling the disease.[8] Parasitological confirmation relies on visualizing the parasite by microscopy and is technically demanding.[18] As fewer cases are found, diagnostic skills gradually disappear. Even the most sensitive parasitological confirmation tests in the hands of experienced technicians most probably will miss up to 20% of cases.[19] In addition there are losses of suspect individuals between screening and confirmation and not all those confirmed are started on treatment. [8]

1.1.1.3 Innovations:

In recent years there have been some major advances in tools for gHAT management and control. Until less than 15 years ago, treatment for the second stage of gHAT still relied on melarsoprol, an extremely toxic arsenic derivative that caused fatal encephalopathy in up to 10% of those treated. [7] Since then Nifurtimox Eflornithine Combination Treatment (NECT) has been introduced, a far less toxic treatment but still requiring IV infusions.[20] Since 2019, a non-toxic oral treatment (fexinidazole) has become available. Fexinidazole is active against both stages of the disease but requires taking treatment for 10 days. Very recently a single dose oral treatment (acoziborole) has successfully completed phase 3 evaluation.[9] Acoziborole (SCYX-7158) is an orally available benzoxaborole-6-carboxamide from a DNDi drug discovery program, which showed high activity *in vivo* in both stages of gHAT in murine models. It also showed low toxicity and strong efficacy in preclinical studies. [21] A first in man study (DNDiOXA001) demonstrated the tolerability and pharmacokinetics (PK) profile for a single oral dose and determined that a 960 mg therapeutic dose in fasted condition provided the desired exposure with a good safety profile. The pivotal study (DNDi-OXA-02-HAT) was a multi-centre, open-label, single arm, prospective trial in adult patients with gHAT with 18 months follow-up. This study showed a high efficacy 95.2% (159/167; CI 95% Jeffreys: 91.2- 97.7) and a very good safety profile. In response to a request from the WHO task force on ‘Integration of new tools into national and global policies’, DNDi have since initiated a double blind study in HAT seropositive non-parasitologically confirmed participants to further assess safety (DNDi-OXA-04- HAT). A pediatric study (DNDi-OXA-05-HAT) is also ongoing and aims to evaluate the PK profile and efficacy in kids from 1-14 years old. This study is expected to be completed before the starting date foreseen for the STROGHAT study.

Another area in which great progress has been achieved is diagnostics, in particular screening tests. In this field too, until recently not much had changed since the introduction of the Card Agglutination Test for Trypanosomiasis (CATT) in the 1980s.[22] The CATT, developed and produced at the Institute of Tropical Medicine (ITM), is a point of care screening test but it does require an electric rotator and a cold chain. It is produced in vials of 50 test units that once opened cannot be kept. Specificity was estimated to be relatively high ($\geq 98\%$) but HAT being a rare disease, the positive predictive value was until now considered too low to justify initiation of treatment, especially given the toxicity of stage 2 treatment.[23] In recent years rapid screening tests (RDTs) based on the lateral

flow principle have been developed and marketed, which are individual, do not require any equipment and do not require refrigeration. Their specificity is probably a little below that of CATT, so until today parasitological confirmation is still the norm before treatment is initiated. [24-26] Current routine data from DRC show very high specificity of CATT, above 99.5%, still making it an attractive option for use in mass screening. (ref. Hasker et al, Current Opinion on infectious diseases)

Equally important innovations in gHAT diagnostics are a highly sensitive and specific inhibition ELISA (iELISA) and new molecular tests. [27, 28] These tests can be applied in a remote laboratory on samples collected in the field, making it possible to confirm gHAT infection on a stored sample. The current diagnostic confirmation tests all rely on visualizing live parasites, which is only possible on fresh samples. Whereas iELISA is an antibody detection test, unable to distinguish between a current infection or an old or aborted infection, the molecular test qTBR, is able to make this distinction, especially if applied on RNA. [29]

In vector control too, major advances have been made. ‘Tiny targets’, a new tool that comes at a fraction of the price of classical tsetse traps, has been shown to dramatically reduce vector density when deployed in a systematic way.[30-32] In this study we will limit ourselves to one intervention however, screen & treat, to avoid confusion over which intervention to attribute any success to.

1.1.1.4 Trends in HAT epidemiology:

In recent years, despite an increase in population screened, annual numbers of gHAT cases detected continue to decrease. In 2019 in the DRC alone, three million were screened but only 613 cases were detected (figure 2). This trend has since continued though figures for 2020 and 2021 may also reflect reduced accessibility to healthcare services as a result of the Covid-19 pandemic.

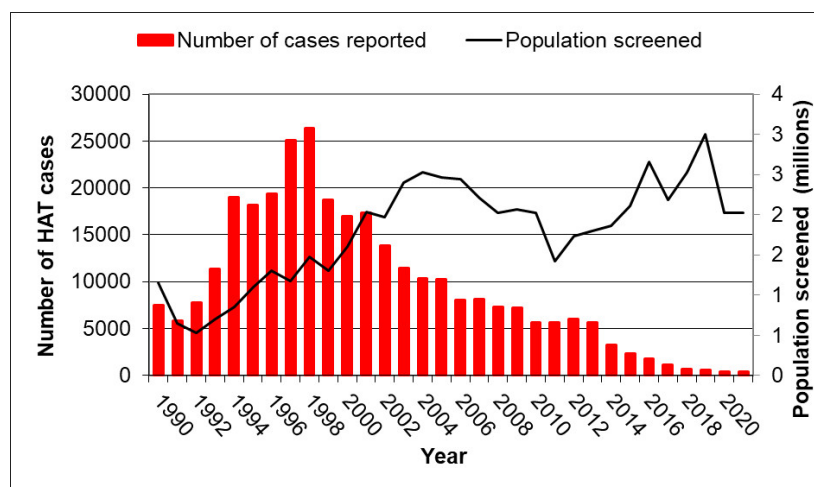


Figure 2: gHAT screening and case notification in the Democratic Republic of the Congo (1990-2021)

An important feature of HAT is its focal distribution. As illustrated in figure 1 above, HAT occurs in distinct foci and as Franco *et al.* reported already in 2016, the surface of areas where HAT was still moderately or highly endemic had halved in less than a decade.[11] By end of 2018 the areas at very high, high and moderate risk had been reduced to 195,000 km² with an annual decline of 40,000 km² in 2017 and 2018.[12] This further illustrates that HAT, and gHAT in particular, can be controlled over the longer term if areas affected are dealt with in a systematic manner and some form of post-elimination surveillance is maintained.

1.1.1.5 Outcomes (efficacy, safety) of completed and number of ongoing clinical studies utilising the same intervention in the same indication (including review of public registers)

Test & treat for elimination of gHAT transmission has been evaluated earlier, between 1985 and 2004, by Simarro *et al.* [33] in an island setting with the drugs then available. They demonstrated feasibility of the concept but treatment options available at that time did not make it feasible for large scale

application in mainland foci. The advent of acoziborole may change this. Acoziborole is a drug belonging to the benzoxaborole class of compounds, characterised by a boron-heterocyclic scaffold, is chemically classified in the BCS class II compound (low solubility, high permeability). Acoziborole is a non-ionized compound (pKa is 9.61), with high lipophilicity (Log D_{ph7.4} = 3.51) and low solubility in aqueous and acidic solution. Preclinical data *in vitro* showed that acoziborole has a minimum inhibitory concentration of 0.6 µg/mL against *T.b. brucei*, which corresponds to approximately two times the IC₅₀ for this parasite species.

Safety studies *in vitro* showed that acoziborole did not interact with a battery of receptors, ligands or enzymes, to a degree that could represent a safety concern. Appropriate genotoxicity studies were performed according to ICH and OECD guidelines. *In vivo* Ames test and *in vitro* mammalian chromosome aberration test did not indicate any potential for genetic toxicity at the highest plasma concentration observed with the highest doses administered in humans (1200mg).

Data from cell-based assays indicate that acoziborole can be readily absorbed across the intestinal epithelium and has a high potential to cross the blood-brain barrier. A radiolabeled excretion balance study in rats and in humans (mass balance study DNDi-OXA-07HAT) indicate that excretion is largely via the faeces and elimination of radiolabelled material is largely complete in both species. Absorption rate in fasted conditions reach 80 to 100% in animals and was calculated to be more than 70% in humans.

A complete package of developmental and reproductive toxicology studies has been completed. No developmental or reproductive hazards were identified. A fertility and early embryonic development revealed no effects of acoziborole on male or female fertility, implantation, embryo-fetal viability or on the number of live embryos up to the highest tested dose level of 25 mg/kg/day. Embryo-fetal development studies revealed no observed adverse effect level (NOAELs) for prenatal developmental toxicity of 15 mg/kg/day in both rats and rabbits. A pre- and postnatal developmental toxicity study in rats revealed no adverse developmental effects of acoziborole up to the highest tested dose level of 25 mg/kg/day. Acoziborole was found to cross the placenta and to be excreted in milk of rats.

A full toxico-kinetic development performed after a single-dose administration to rats, showed a maximum tolerated dose of 200 mg/kg. In the 28-day repeated-dose toxicology studies, the NOAEL was set at 15 mg/kg/day for both rats and dogs.

In order to fully cover the length of time patients will be exposed, toxicity studies in addition to the prior 28-day studies were conducted. In these GLP studies, rats and dogs were treated for 13 weeks followed by a 6-week recovery period. The rat was the most sensitive species and is to be considered when estimating the therapeutic index. The effects observed in the 13-week toxicity study in rats, in particular the spermatic granulomas, were related to the multiple and repeated administrations for 13 weeks, i.e. 65-66 doses in total. As compared to the overall exposure in patients, the cumulative exposure observed in rats after 13 weeks of treatment, was 2-fold higher than the NOAEL and 7-fold higher at the dose level where the findings in the epididymis were observed.

The preclinical data have been reviewed by an independent toxicologist and no further preclinical studies were requested, especially no juvenal toxicological study is needed.

A number of studies in humans have now been completed. These include the following:

1) **DNDi-OXA-01-HAT**, A Phase-I randomized, double-blind, placebo-controlled sequential study, was conducted in healthy African male volunteers aged 18 to 45 years to assess the safety, tolerability, pharmacokinetics and pharmacodynamics of acoziborole after single oral ascending doses. A total of 128 subjects were included, 102 of whom received acoziborole at doses from 20 to 1200 mg, and 26 of whom received placebo.

PK results showed that acoziborole was quite rapidly absorbed and that levels remained steady for at least 96 hours. Metabolism of acoziborole was very slow leading to an extended half-life of around 400 hours. The pharmacologically active exposure was reached with the 960-mg dose with a

probability of 90% to be above the threshold of concentration in the CSF, i.e. AUC_{fu} of 5.8 µg•h/mL. The highest dose administered in the study was 1200 mg. No safety signals were observed. The exposure corresponding to the 72-96h period was 63% of the free Cav in rats and 37% of the free Cav in dogs, indicating a safety margin of approximately 1.5 to 3-fold in relation to exposure at the NOAEL. Furthermore, inter subject variability was low at around 20%.

Overall, the most frequent acoziborole-related adverse events (AEs) were GI disorders (abdominal pain, nausea, constipation, and vomiting), laboratory abnormalities (CPK increase, transaminase increase and thyroid hormone fluctuations) and nervous system disorders (mainly headache), observed at various doses of acoziborole. Adequate monitoring of possible emerging adverse experiences, including extensive biological parameters i.e.; biochemistry, haematology, coagulation, intravascular bleeding tests, liver function tests, pancreatic function tests, thyroid function tests, lipid profile, clinical disturbances as well as regular physical examination, in addition to PK and electrocardiograms were performed. No dose-related adverse effects and no abnormal findings for physical examinations or vital signs were observed.

2) **DNDi-OXA-03-HAT (NCT04270981)** A Phase-I single-center, open-label, single-group, non-randomized, single oral dose study, was conducted in 6 healthy Caucasian male subjects to assess mass balance recovery, PK, metabolite profile and metabolite identification. The PK profile in healthy subjects was similar to that in the Phase-I study. The radiolabelled drug was totally recovered at 119 days after dosing, and 30% was detected after 72 hours, corresponding to an absorption fraction (fa) of 70% of the administered dose. The parent compound present in plasma and faeces represented the vast majority of the radioactivity, i.e. 95.0% and 66.0%, respectively, and only 7.6% of the radioactivity was recovered in urine. Faecal excretion represented the predominant route of excretion of drug-related material. The same metabolites identified in animals were identified in humans.

3) **DNDi-OXA-02-HAT (NCT03087955)**, A phase II/III, multicenter, open-label, prospective study, to assess the efficacy and safety of a single dose of 960 mg of acoziborole under fasting conditions in patients with HAT due to *Trypanosoma brucei gambiense* (gHAT) was conducted in 208 male and female patients of sub-Saharan African origin, over 15 years of age, in the Democratic Republic of the Congo (DRC) and Guinea. A total of 27 serious adverse events (SAEs) were reported in 21 patients exposed to acoziborole, among which 4 fatal cases. All SAEs were assessed as not related to acoziborole by the investigators and the sponsor. The most frequently reported SAEs were infections and infestations or psychiatric or central nervous system (CNS) disorders. Seven cases of pregnancy were reported during the study period, 6 of which involved exposure during pregnancy, no adverse events were observed during pregnancy duration. There were 4 neonatal deaths but none were related to acoziborole, three related to difficulties during (unassisted) delivery and one to accidental suffocation while breastfeeding.

The treatment success rate at 18 months in late-stage HAT patients was 95.2% (159/167; CI 95%, 91.2 - 97.7) for the mITT set, and 98.1% (159/162; CI 95%, 95.1 - 99.5) for the evaluable patients. Efficacy results at M12 and M18 were the same and compare favourably with rates observed in historical NECT studies (estimated to 94.1% [CI 95%, 91.4; 96.2]). In early- and intermediate-stage HAT patients, the success rate at Month 18 in the mITT set was 100% (95% CI, 94.1 – 100).

At 18 months, 8 patients were considered as treatment failures (3 relapses treated with rescue therapy; 4 deaths unrelated to efficacy, safety, or disease evolution; and 1 loss to follow-up). The evaluable patients set, which is closely aligned with WHO criteria, excluded 5 patients from the ITT (the 4 deaths and 1 loss to follow-up) leaving only 3 patients considered as failures. Figure 3 below shows a comparison of DNDi-OXA-02-HAT to other recent studies on treatment of gHAT.

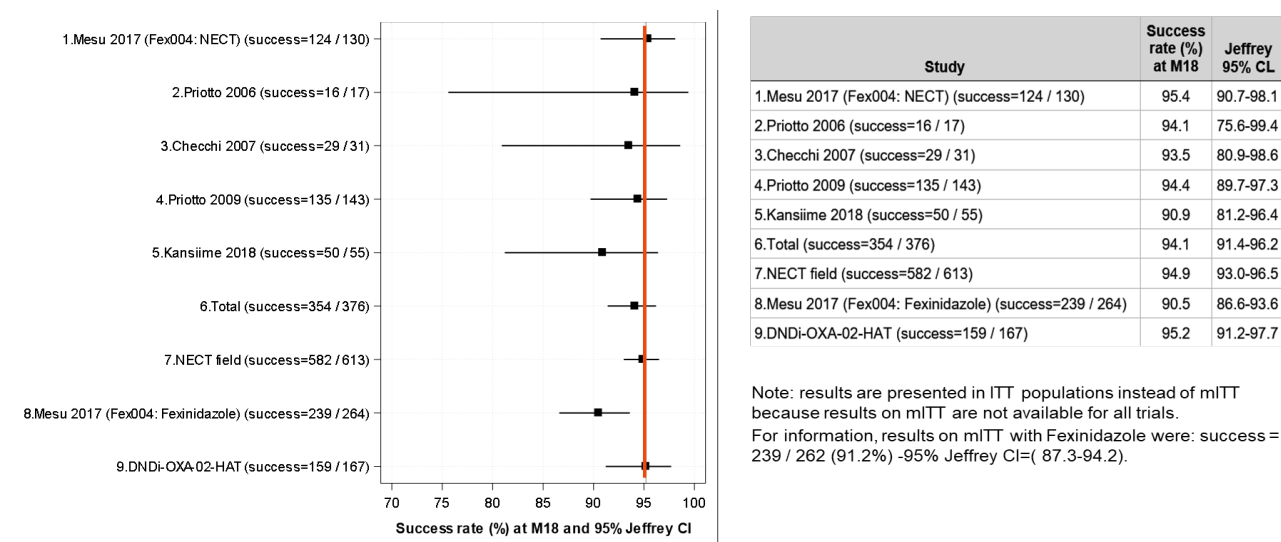


Figure 3: Results of studies on effectiveness of recent gHAT treatments

A number of trials are currently underway, these include:

1) **DNDI-OXA-04-HAT (NCT05256017)**, a phase II/III Safety and tolerability study in gHAT seropositive non-parasitologically confirmed subjects using a multicenter randomized double-blind placebo-controlled design. 1200 participants (900 active and 300 placebo) are planned to be enrolled. End of study is expected in December 2022. The study is running in DRC (5 sites) and in Guinea 1 site + 2 satellites sites)

Primary objective:

- To assess the safety and tolerability of a single dose of acoziborole compared with placebo during a follow-up period of 4 months in gHAT seropositive individuals who are not parasitologically confirmed.

Secondary objectives:

- To assess safety of acoziborole versus placebo in:
 - Haemato-biochemistry
 - Electrocardiogram data at Day 5 (double delta versus baseline and time, age and sex matched comparison)
- To assess blood PK at Day 5 and Month 1:
An exploratory trypan-skin sub-study in 725 of the participants is included to evaluate the prevalence of extravascular dermal *T.b gambiense* and compare different methods of molecular biological tests

The recruitment has started in December 2021, 20 participants / month/site are include so far and it is forecast to end by December 2022. The complete analysis of the results is foreseen to be available Q2, 2023. No unexpected adverse events were observed so far.

2) **DNDI-OXA-05-HAT (NCT05433350)**, a pharmacokinetics, efficacy, safety, and tolerability study of a single dose of acoziborole under fasting conditions in paediatric patients from 1 to 14 years of age and with human African trypanosomiasis due to *T.b.gambiense*: a multicentre, open-label study. This study is supported by a grant of EDCTP2 (Horizon 2020 EU funding). The study is running in 5 sites in DRC and 1 site in Guinea, 35 children with gHAT are planned to be included. The design is an Open-label, non-randomised, single arm, multicentre, phase II/III study. Participants will be hospitalised for 15 days, followed by an ambulatory follow-up at 3, 6 and 12 months. The doses have been chosen based on modelling of adult exposure, this modelling will be adapted with the first observed PK results in the elderly kids and then weight based dose will be given to the lowest body weight kids over 1 year old. The recruitment is foreseen to end in Q4 of 2023.

1.1.1.6 Level of evidence related to the mechanism of action of the intervention in the planned clinical study population

The current gHAT control strategy is based on active screening of villages at risk by mobile screening teams. All villages from which gHAT cases were reported are screened for three consecutive years until no further cases are reported. They are then screened once more, five years after the last case was reported. If no further cases are found, transmission is assumed to have been interrupted. This strategy has had a major impact as can be seen from figure 2 above. However Robays *et al.* estimated that up to 50% of prevalent cases are not detected or not cured, with major losses occurring in the parasitological confirmation step.[8] Fear of the lumbar puncture and of toxicity of treatment, in particular associated with melarsoprol and still very prevalent in particular in the older population, are other important barriers. [34, 35]. Even if up to 50% of prevalent gHAT cases were not treated, the epidemiological data show that the disease is on the decline.[12] This may however be insufficient to achieve complete elimination of transmission. Simarro *et al.* have demonstrated in an island setting that it is possible to achieve elimination of transmission by treating not just the parasitologically confirmed but all serological suspects identified. [36] At that time treatment options were still limited to pentamidine for stage 1 gHAT and melarsoprol for stage 2 gHAT, making the dreaded lumbar puncture mandatory. We now have a drug available that treats both stages, is fully oral and can be provided as a single dose. We therefore have good reasons to assume that with this kind of tool, elimination of transmission is also possible in a mainland focus.

1.2 Objectives

Primary objective:

- To evaluate if a ‘screen and treat’ strategy can lead to interruption of gHAT transmission in a mainland focus

Secondary objectives:

- To provide further evidence for safety of acoziborole in serological gHAT suspects
- To provide an accurate cost estimate for a screen & treat and treat strategy

1.3 Characteristics of the study population

To select our study population we have received from WHO an update on their risk analysis mentioned in chapter 1.2.1 based on the data for 2019, the last year before the start of the Covid-19 pandemic. During 2019 screening coverage in DRC was at an all-time high of over three million people, as shown in figure 2 above.

Figure 4 below shows the result of this analysis for DRC, i.e. the areas at risk for gHAT. We identified 20 potential intervention areas with an annual risk of infection above 1 per 10,000. Among those we selected an area that is accessible and that is surrounded by areas of low to negligible risk. Security concerns were also taken into account. Thus we opted for the area outlined by the blue dotted line in figure 4 below, in the province of Equateur North.

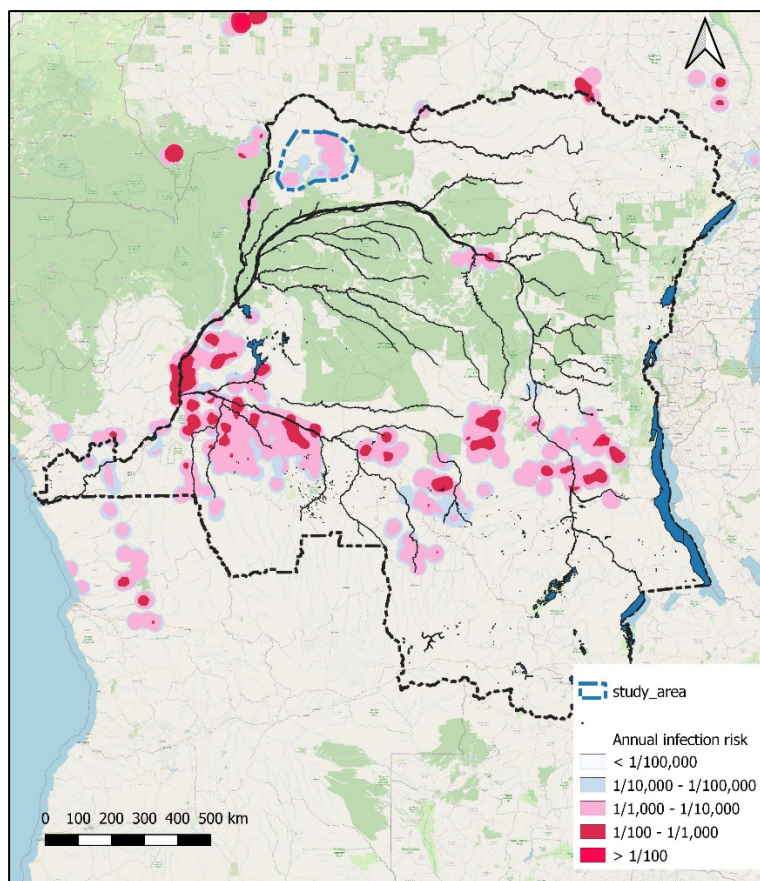


Figure 4: Moderate to high gHAT risk clusters in DRC

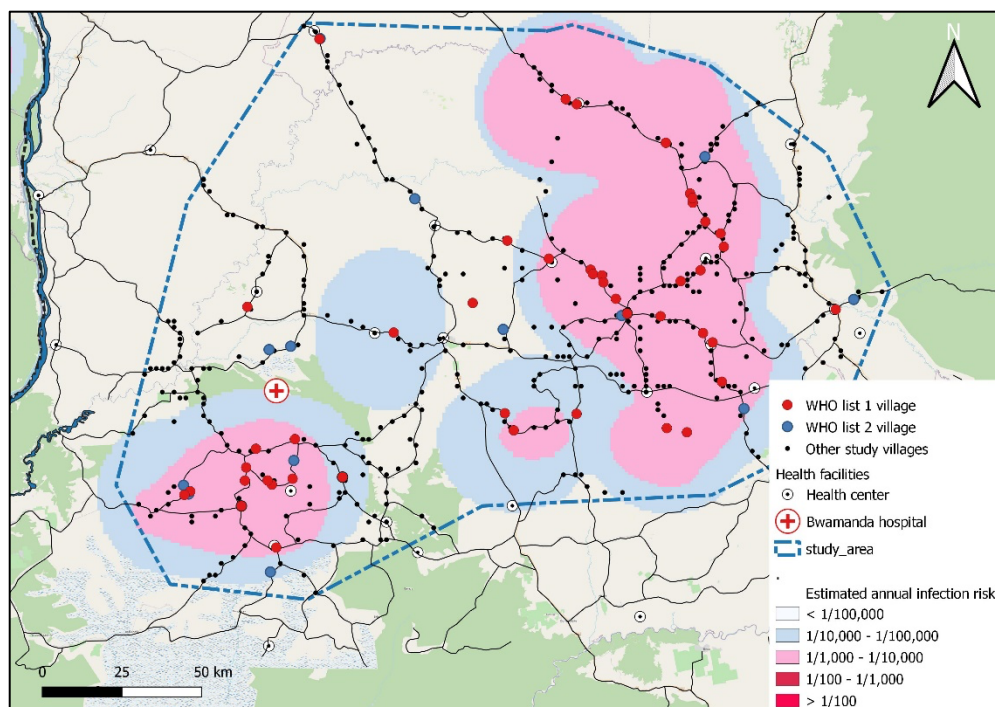


Figure 5: The intended study site in Equateur North province, DRC

The study area (Figure 5) is currently made up of two moderate risk clusters for gHAT. The total population of the 403 villages of the study area is estimated to be between 2 and 2.5 million, based on

data from Landsat and WorldPop.[37, 38]. During the first study year, all villages in the study area will be mapped and more accurate population estimates will be obtained. These estimates will be used to draw a cluster random sample to assess gHAT prevalence in the final study year.

Of the 63 villages identified to be screened by mobile teams (according the national strategy), gHAT cases have been reported over the last three years in 48 villages (i.e. list 1 of WHO) with an estimated population of approximately 130,000, the 15 villages from which gHAT cases have been reported within the last five years but not within the last three years (list 2 of WHO) have an estimated population of approximately 45,000. These two combined will be the population to be screened by mobile teams in the first 3 years of the study. Villages from list 2 where no further gHAT cases are detected, will be removed the next year. Villages on list 1 that had no cases for three years will be moved to list 2. For the purpose of the study a gHAT case is defined as a person testing positive by a molecular test or parasitologically confirmed.

Screening at the villages will be done door-to-door to maximize participation and coverage of all sex and age groups. We will include both genders and all ages for gHAT screening. As soon as further data from the DNDI-HAT-OXA05 study are available (expected by end of 2023) and approval from the DSMB has been obtained, children aged 1-14 years will also be eligible for acoziborole. Former HAT patients will be excluded as serology is known to remain positive for years. All individuals who do not agree or are not eligible to be included in the study will be managed according to national guidelines. Severely ill patients in need of hospitalization will be referred to the hospital and managed according to national guidelines. Women of childbearing age will be offered pregnancy testing prior to enrolment. If positive they will still be eligible for enrolment but follow-up will be closer; as per DNDI clinical SOPs, pregnancy will be followed every 3 months and to every mother it will be proposed to be hospitalized at end of pregnancy to ensure assistance at delivery. New-born exposed *in utero* will be followed during 2 years after birth.

Throughout the study screen & treat will also be offered in the 30 WHO sentinel sites¹. They will screen by an RDT and if positive collect a blood specimen (plus a lymph node aspirate if applicable) and presumptively treat with acoziborole. The selection of patients eligible for RDT screening will be based on the WHO guidelines for gHAT sentinel sites. Whenever a case is confirmed, the village of residence of this case will be added to list 1 (if the village is not yet included).

1.3.1 Details on sample size and power calculation

For active screening by mobile teams, we expect a target population of 175,000 in year one, reducing to 130,000 in year 3. Assuming that 85% will participate and that 1% will test positive in the screening test, we expect to treat 1,500 individuals in year one and approximately 1000 more during year two and three. This complementary sample size of 2500 participants added to the DNDi-OXA-04-HAT-study (900 active +300 placebo) will provide over 90% power to detect even uncommon adverse events with a frequency of 1 per 1,000. A meta analysis is planned to include all AEs collected after acoziborole administration.

¹ WHO sentinel sites in the study area are HGR Bwamanda, HGR Gemena, HGR Karawa, HGR Tandala, CSR Botetenzza, CSR Bondama, HGR Bokonzi, HGR Bangabola, CH Yakamba, HGR Bogose Nubea, CSR Takaya, HGR Loko, HGR Businga, CSR Movoto Bozene, CSR Gbatikombo, HGR Kungu, HGR Boto, HGR Budjala, HGR Bominenge, CSR Bodeme, CSR Bowara, CSR Musa, CSR Engalango, CSR Bogbakusa, CSR Mbonza III, CSR Boso-Modjebo, CS Cité I, CSR Boyazamo I, HGR Bosobolo, HGR Gwaka.

During the 4th year we will screen all those living in villages that were on list 1 or 2 at the start of the study or added later ($\pm 150,000$), plus an additional 150,000 from the remaining villages, i.e. not on list 1 or 2. The villages have an estimated population between two and two and a half million. As population data differ, more accurate population data per village will be gathered during the first study year, and in year 4 we will draw a cluster random sample based on updated population data. This screening will be focused on adults, who are known to be at an above average of gHAT infection.[39] If in this population of 300,000 no further cases are found, assuming a design effect of 2, with 95% certainty the true prevalence will be below 1 per 50,000.

1.4 Design of the clinical study (controlled / uncontrolled; randomised; open / blinded; parallel group / cross over / other; please justify the appropriateness of the selected design)

The ideal comparison would be between an area in which standard of care is implemented and an area in which screen & treat is implemented. Given the already very low incidence levels of gHAT, demonstrating a difference between such study arms would require a sample size that would not be realistically achievable. We therefore opted for a quasi-experimental design aimed at demonstrating a prevalence of below 1 per 50,000 after three years of intervention. We assume that at such levels of prevalence R0 will be below 1 and the disease will head towards extinction. The known endemic villages at the start of the study will be screened in their entirety at the end of the study, allowing a very precise estimate of gHAT prevalence after the intervention. Thus we will be able to arrive at a robust assessment in the villages screened by mobile teams of whether interruption of transmission has been achieved.

For the other villages, where no cases were reported through the health system prior to the start of the study, we will have a cluster random sample of 150,000, deliberately over-representing those most at risk, i.e. adults. This should provide an indication of ongoing transmission in the wider study area.

Nested in this epidemiological study will be a phase 3b trial on safety of acoziborole. This study will be an open non-randomised single arm group to complement the results of the DNDI-OXA-04-HAT which is a randomised double blind versus placebo study. Study participants treated with acoziborole will be followed up actively until five days post treatment for adverse events with a final assessment of treatment outcome at between 3 to 4 months – post treatment.

1.5 Type of intervention (medicinal product / advanced therapy medicinal product / medical device / in vitro diagnostic medical device / surgical or other invasive procedure / other medical intervention, including, e.g., counselling)

1.5.1 Study treatments

The Investigational Product (IP) is acoziborole 320 mg. The IP will be provided by DNDi, 15 Chemin Camille-Vidart, 1202 Geneva, Switzerland.

1.5.2 Doses and treatment regimens

For adults three tablets of 320mg of acoziborole will be provided in a single intake by the oral route. For the kids below 30kg, adjustment will be made according to weight bands- in accordance with the paediatric study).

Modelling of the dose in kids showed 3 body weight bands:

10-20 kg body weight: 320 mg single dose (pediatric formulation)

20-30kg body weight: 480 mg single dose (pediatric formulation)

30-40kg body weight: 640 mg single dose (or 2 adults tablets of 320mg)

The pediatric study (1-14 years old kids) is aimed to collect observed PK sampling to confirm or adjust this modelling.

1.5.3 Study treatments labelling, packaging

The IP will be packaged in aluminium-aluminium blister packs. Each pack will contain the number of tablets required for one administration (three tablets) and will be packaged in an individual treatment kit of each subject. The packaging will be sponsored by BMGF.

The labelling on the treatment kit will display the following information:

- Name of the Sponsor* and name and contact details of the Coordinating or Principal Investigator
- Study number*
- Name* and dose strength of the IP
- Dosage form*, route of administration*, and number of dose units
- Instructions for use
- The statements “for clinical study use only” and “keep out of the reach of children”
- Batch number* and number of treatment kit*
- Expiry date and storage conditions

The information items marked with an asterisk (*) will also be displayed on the immediate packaging of the IP. Information on acoziborole will be provided in the Investigator’s Brochure, attached to the protocol submitted to the Competent Authorities.

1.5.4 Accountability

The study-specific forms must be used for accountability of the IP. Appropriate records concerning receipt, use, return, loss, or other disposition of the IP must be maintained by the Investigators at the Provincial Coordination of national sleeping sickness control program (“Programme National de Lutte contre la Trypanosomiase humaine Africaine” PNLTHA) or study centres or by their delegates, under the supervision of the Principal Investigator. In addition, the study monitors must check accountability of the IP during their on-site monitoring visits. The IP will be stored in the DNDi Kinshasa pharmacy and regularly shipped to the Provincial Coordination of PNLTHA for secondary storage. In those storage centres, the IP must be stored in a locked room or a locked cabinet with access limited to the person in charge of the pharmacy or authorised study personnel. The IP must not be used for purposes other than the present protocol. The Investigator and the study personnel may not, under any circumstances, provide other investigators or healthcare services with the IP attributed to their study centre or allow IP to be used other than as described in this protocol without prior written approval from DNDi.

1.5.5 Storage

The IP must be transported and stored at a temperature not exceeding 30°C. It is recommended to store IP at room temperature. The IP should not be used beyond the expiry date indicated on the label. The storage conditions, including the temperature, at the Provincial Coordination of PNLTHA or study centres must be monitored by the study personnel and appropriate records must be available. Any temperature excursion should be recorded and reported to DNDi.

1.6 Description and timing of study procedures

Active and passive screening is part of the routine HAT control program activity in DRC, participants will only be enrolled once they test positive in serological screening. Those accepting to participate will have a venous blood sample (and if typical lymph glands are present, a sample from a lymph gland puncture) taken for remote diagnostic confirmation, after which they will be visited in their homes by a study physician who will provide the study treatment, i.e. 3 tablets of acoziborole for an adult, taken as a single dose. The study physician will pay a second home visit 3-5 days after ingestion of the study drug to inquire about adverse events. There will be a final visit three months after treatment to assess whether there are (still) any clinical suspicions of gHAT. At this occasion for participants that were confirmed as gHAT cases by molecular test, a new venous blood and if applicable lymph node sample will be collected and stored at -20°C to be tested for the presence of RNA of *T.b. gambiense*, indicative of an ongoing active infection.

Throughout the study period samples collected will be kept at the Provincial Coordination of PNLTHA or study centres and shipped to the reference laboratory at regular intervals. There they will be screened first with an indirect ELISA or inhibition ELISA, those testing positive will be tested by immune trypanolysis (the current gold standard for serology) and molecular tests, with a focus on

detection of DNA. From all confirmed gHAT cases, i.e. positive on molecular testing, a second sample will be collected three months after treatment to be tested for the presence of RNA.

Persons screened and found seropositive but not agreeing to participate or meeting the exclusion criteria of the study (e.g. children 1-14 years of age (until availability of data allowing to relax this exclusion criterion) or severely ill persons) will be managed according to the national guidelines for gHAT control in DRC.

This will allow us to draw a cluster random sample during the fourth study year in which the number of participants to be sampled from the villages that were not actively screened (not on list 1 and 2) will be proportional to their estimated population sizes. Table 2 below provides an overview of the main study activities.

Table 2: Overview of study activities

Study year	Villages list 1	Villages list 2	Other villages	Laboratory
1	Door-to-door active screen & treat Monitoring (S)AEs/ Rx outcome	Door-to-door active screen & treat Monitoring (S)AEs/ Rx outcome	Passive screen & treat, Monitoring (S)AEs/ Rx outcome, mapping + population estimate	Processing of samples
2	Door-to-door active screen & treat Monitoring (S)AEs/ Rx outcome	Door-to-door active screen & treat Monitoring (S)AEs/Rx outcome	Passive screen & treat Monitoring (S)AEs/Rx outcome	Processing of samples
3	Door-to-door active screen & treat Monitoring (S)AEs/ Rx outcome	Door-to-door active screen & treat Monitoring (S)AEs/Rx outcome	Passive screen & treat Monitoring (S)AEs/Rx outcome	Processing of samples
4	Door-to-door active screening	Door-to-door active screening	Active screening of random sample	Processing of samples
5	Final analysis of all study data			

As stated above, gHAT screening offered is part of the normal routine of the PNLTHA. Persons testing positive will have the choice between participating in the study or being treated in accordance with standard of care. Table 3 below shows more detail on procedures to be followed if a person tests positive at the time of gHAT screening and provides informed consent for being enrolled in the study.

Table 3: Overview of study activities for serological gHAT suspects

	Screening	Dosing by investigator	Post dosing period					
			D1	D2	D3	D4	D5	M3-4
(Serological test RDT)	X ¹							
Informed consent	X							
Blood sample to be stored for complementary analysis	X							
Pregnancy test ⁴	X							X
Demographics	X							
Medical History	X							
Concomitant medications record	X	X		X	X	X	X	X
Clinical examination	X						X	X ²
Treatment		X						
Collection of AEs	X	X		X	X	X	X	X

1 Persons screened and found seropositive but not agreeing to participate or meeting the exclusion criteria for acoziborole (e.g. children below 5 years of age or severely ill persons) will be managed according to the national guidelines for gHAT in DRC

2 Of all participants that were confirmed as gHAT cases by molecular test, a new venous blood sample will be collected and stored at -20°C to be tested for the presence of RNA of *T.b. gambiense*, indicative of an ongoing active infection.

2 Preparedness status

2.1 Development of the clinical study protocol

2.1.1 Scientific advice from regulatory and health technology assessment bodies

Before any study subject is enrolled we will seek ethical approval, in line with regulations for a clinical study (*phase 3b trial on safety*) and for an epidemiological study to a relevant ethics committee and as applicable to the Regulatory Authorities in DRC. Implementation will only start once this approval has been obtained.

2.1.2 Clinical efficacy, safety, and methodological guidelines (including guidelines on statistics)

Detailed information on earlier and ongoing clinical trials on acoziborole was presented in 1.2.1.1 above. In brief, clinical efficacy and safety of the study drug acoziborole have been assessed for gHAT patients in a trial that was recently completed (DNDi-OXA-02-HAT, NCT03087955). The 18 months intention-to-treat success rates were respectively 95.2% for meningo-encephalitic stage HAT (n=167) and 100% for early/intermediate stage HAT (n=41, white blood cell count $\leq 20/\mu\text{l}$ and no trypanosomes in CSF). Acoziborole showed a good safety profile, with no attributable serious or severe treatment emergent adverse events (n= 208). Mild and moderate treatment emergent adverse events occurred in 14% of patients, mainly on day 1 (10.8%) and day 2 (2.9%), while no significant biochemistry and hematology changes were observed. Despite taking precautions, pregnancies occurred, but no particular problems related to potential drug exposure of the foetus were observed. EDCTP2 is currently financing a clinical trial on pediatric treatment of gHAT with acoziborole (ACOZI-KIDS or DNDi-OXA-05-HAT, EDCTP2/RIA2019PD-2890). Another ongoing trial (DNDi-OXA-04-HAT) conducted by DNDi assesses safety of acoziborole in serological gHAT suspects who remain negative in the microscopic parasitological confirmation. Preliminary results of this trial are expected in the 4th quarter of 2022.

The primary outcome would be based on 2 indicators. On the one hand on the before-after comparison of prevalence in the 63 villages, and on the other hand on whether or not there is indication of interruption of transmission of *T.b. gambiense* based on data from a cluster random sample. We will calculate the prevalence of *T.b. gambiense* infection in the villages that have been screened during the three intervention years (list 1 and 2 of WHO) as well as in the villages that were not targeted with active screening but are part of the study area outlined in figure 5 and in all villages across the study area. We will use survey statistics in Stata (StataCorp, 4905 Lakeway Drive, College Station, Texas 77845 USA) to calculate proportions of infected individuals among all sampled. As sampling weight we will use the inverse of the probability of being sampled, i.e. the estimated village population divided by the number of persons sampled for that village.

2.1.3 Involvement of citizens / patients, carers in drawing up the clinical study protocol

The PNLTHA has a long history of conducting population screening for gHAT in the villages targeted. Public awareness campaigns making use of village health workers are part of their strategy. Before starting the trial an acceptability study will be conducted by social scientists who will explore acceptability of the study procedures and factors associated with participation. Activities to maximize participation of population in the villages screened by mobile teams will be launched. Screening activities at village level will be preceded by an information/education/communication campaign to

inform the population on the changes in screening procedures. Village health workers (*Relais Communautaires* – RECO) will be involved throughout the study.

2.2 Regulatory intelligence to ensure timely regulatory approval and ethics clearance of the clinical study in all jurisdictions where its implementation is planned Please provide information on the following regulatory and ethics aspects:

2.2.1 How the consortium will ensure access to regulatory expertise necessary to get advice on, and management of, regulatory affairs activities in all concerned jurisdictions?

DNDI has ample experience in conducting clinical trials, including initiation, supervision and monitoring in DRC.

2.3 How the scientific and operational governance of the clinical study will be ensured?

2.3.1 Please give details about the sponsor(s) (name, type of entity, seat or country of residence).

The sponsor of this trial will DNDi, who will ensure the following responsibilities:

- Monitoring activities in the field;
- Preparing and organizing the submission dossier for the regulatory authorities (including annual reports and end of study notifications);
- Organizing regular Trial Management Group (TMG) meetings during the whole project ;
- Data Management activities (including design, system validation, data management plan, data review and data cleaning) ;
- Statistical activities (including statistical analysis plan, interim analysis, final analysis)
- DNDI will be accountable for the safety follow-up activities (including SAE reports receipt and follow-up, organizing regular DSMB meetings, reporting to IRB/ECs and regulatory authorities).

2.3.2 Please describe the composition, the role and the functioning of the planned board(s), governing bodies

For STROGHAT we will establish an external scientific advisory board in which WHO has already agreed to participate. We will also invite some individual HAT experts and if possible experts from other research institutes/ universities. It needs to be borne in mind though that most research institutes and organizations involved in HAT control are already part of the consortium.

For the clinical trial (*phase 3b trial on safety of acoziborole*), a Trial Management Group will be set-up (including at least the sponsor coordinating investigator, sponsor laboratory manager, data manager, statistician, monitor, project manager and principal investigators from both sites) which will meet at regular intervals (typically monthly) to discuss the daily management of the trial. A Data Safety Monitoring Board will be set-up with external HAT experts who will review the safety within the trial (including review of the SAE reports and regular DSMB meetings).

3 Operational feasibility

3.1 Please describe how the availability of the intervention(s) (including comparators) is secured throughout the entire implementation phase (give details on manufacturing, packaging / labelling operations, storage, logistical, import/export issues, etc.)

The intervention will require annual screening of 150,000 individuals over a 3-year period, followed by a final screening round of a population of 300,000, spread out over 303 villages. An estimated

2,500 subjects will be treated with acoziborole. Apart from human resources and equipment, this requires availability of screening tests and acoziborole. The main screening test used will be CATT for mobile teams and the RDT (native HAT Sero K-SeT) in passive screening.

Considering a screening capacity of 60,000 persons per year, three mobile teams of the PNLTHA will conduct active screening in the eligible villages (n=63). These teams will be expanded by 1 medical doctor and 1 nurse, and 1 additional means of transportation to ensure the required task on top of the routine activities (e.g. informed consent, sampling, treatment, follow-up, ...).

The number of serological suspects to be treated is higher than numbers of HAT cases currently treated in DRC but since this is a single dose oral treatment, it will be logistically much easier than all other treatments previously used. Acoziborole will be provided through DNDi who is part of the consortium.

ITM and DNDI have a registered office in Kinshasa, DRC, exempted from import taxes. The Ministry of Health of DRC are involved through their HAT control program, the PNLTHA.

3.2 Please describe how the study population will be recruited

The study population will be recruited during routine gHAT screening activities of the national HAT control program, the PNLTHA. Three mobile teams of the PNLTHA will conduct active screening in eligible villages (those on list 1 and 2 of WHO) and anyone testing positive to the serological screening test will be given the opportunity to be enrolled in the study. The MD will decide whether the sero-suspect candidate can participate in the study. An informed consent will be obtained on the spot. From all participants, a venous blood sample (plus lymph node aspirate if applicable) will be taken for remote post hoc testing in Kinshasa, acoziborole will be administered and follow up for adverse events will be ensured. Participants with signs of severe gHAT or other intercurrent symptoms of clinical severity will be managed according to the national guidelines of gHAT in DRC.

In each village a RECO will be involved to sensitize the population before the arrival of the team and to guide the study physician to the home of the participant for follow-up.

The screen & treat strategy will be applied in the fixed health facilities, following similar procedures as in the mobile teams. However the decision to apply an RDT will be based on the WHO guidelines for gHAT sentinel sites. These facilities will be provided with sufficient quantities of RDTs. Anyone testing positive and residing within the study area will be offered the option of being enrolled in the study, in which case an informed consent will be signed and a venous blood sample (plus lymph node aspirate if applicable) will be collected and stored. The study team will be contacted and will ensure that the sample is forwarded to the laboratory and the participant is visited at home for provision of the study drug, post-treatment observation during 5 days, and further follow up.

During the final survey in the 4th year, villages that were originally in list 1 or 2 or were added later during the study, will be screened exhaustively, i.e. the whole population and all ages. From the remaining villages a random sample will be drawn, proportional to the estimated population sizes, using probability proportional to size.

All recruitment will take place in one geographically continuous area in the province of Equateur North in DRC. Samples collected will be processed at the national reference laboratory in Kinshasa, the INRB.

3.2.1 Will recruitment of the study population be of competitive nature between the clinical sites? (Please describe how underperformance of individual clinical sites in recruitment will be managed.)

There will be only one site, serviced by three mobile screening teams and a number of fixed health facilities. Enrolment will be exhaustive, all serological suspects identified during the study period are eligible.

3.2.2 What evidence supports the ability of the individual clinical sites to recruit the required number of study participants within the planned timeline (e.g. documented performance in previous clinical studies of similar complexity targeting very similar study population)?

PNLTHA has the ability to screen large populations. Even during the years of the Covid-19 pandemic, the PNLTHA still managed to screen up to two million persons per year.

The villages targeted are already being covered by mobile teams of the PNLTHA for active and passive gHAT screening. Normally the mobile screening teams provide parasitological confirmation on the spot and even perform the lumbar puncture if required. This will no longer be necessary in this study, the teams can focus on screening only and recruit those testing positive into the study. On average a mobile team screens approximately 300 persons a day, of which approximately 1% are expected to test positive. Thus each mobile team will recruit approximately 3 participants per day, which is a very small number. Three mobile teams will work in parallel, and a number of fixed health facilities will also recruit.

3.3 Please describe what additional supply (e.g. an electronic device for remote data capture, a specific instrument for administering the investigational product, etc.) is necessary to carry out the required study procedures and how this supply will be made available to the clinical sites

The routine screening data will be captured via the TRYPELIM digital tool, a custom designed mobile app for data collection, which is routinely used by the PNLTHA. This is an application for data collection and dashboard. It also allows to take pictures and videos which can be used for quality assurance at a later stage.

For those enrolled in the clinical study we will make a separate data collection tool in MEDRIOa GCP approved software previously used in DNDi studies, that allows for offline data collection and for recording longitudinal data. To ensure adequate follow-up also after three months, we will record the geographic coordinates of the household of each participant enrolled for treatment. Data collected both by the routine system and by MEDRIO are uploaded to a secure server whenever internet access is available. If required a VSAT server will be installed at the Bwamanda hospital, centrally located in the area.

3.4 Please provide plans on data management aspects (data standards, type of data capture, verification of data, central data collection, cleaning, analysis, reporting, security)

Data management of the TRYPELIM digital tool will be managed by PNLTHA. These data will include data of all persons screened by the mobile teams, both sero-negatives and sero-positive suspects. All data of the phase 3b trial on safety of acoziborole will be managed by DNDi, on a server based in Kenya, where DNDi manages their other clinical studies.

A Data Management Plan will be prepared, with following essential aspects:

- Data collection, handling and retention

Household-level (geographic coordinates) and participant-level data (demographic, health and medical data) will be collected via electronic questionnaires on mobile devices (smartphones, tablets and/or laptops) into the study database, by using MEDRIO. In addition, data of all participants (sero-positives and sero-negatives) will be recorded through the routine data system.

The MEDRIO Electronic Data Capture software has been successfully used by DNDi in previous clinical trials. It is a secure web application as well as mobile application that can be used to collect, clean and manage subjects' information in accordance with various standards and applicable regulations (GDPR, ISO27001, GCP, CFR 21 part 11, HIPAA). Furthermore, the software features amongst others a query management system and capabilities for importing and exporting data in various forms. MEDRIO allows for offline data collection in remote places. When connecting to the

internet or cellular networks, data from the mobile device can be synchronized with the study server where the study data will be securely retained.

Paper questionnaires might be used if deemed necessary (e.g. as a backup when having issues with the electronic device). Only data defined by the study protocol will be collected.

Questionnaires will be tested and validated before the household visits start. All study personnel will be trained on informed consent and study procedures, data collection and entry before the study start.

Edit checks and branching logic, programmed onto the electronic forms will validate at the point of data entry and support quality data. In addition, the data will be reviewed and monitored during the course of the study by various study collaborators. Data cleaning and querying of inconsistent, inaccurate and missing data will be done following a Data Validation Plan to achieve a timely database lock.

- Data security and confidentiality

The DNDi system includes a robust security architecture including encryption, firewalls, antivirus software and controlled user access (smartphone/tablet authentication, username, personal password and authorized user role). A list with all users who have access to the system will be kept updated during the study. A daily backup of data will be provided at the study server.

Both the sponsor and the study site will see to it that the necessary measures are taken to ensure that all data management documents (study Data Management Plan, procedures, completed questionnaires) and IT devices and equipment (mobile devices, computers, server) are kept secure (closed cupboards, closed and/or badge controlled offices and/or rooms).

Private data of the study participants will be handled confidentially. Information such as the participant name or any other data which could lead to the identification of the participant will not be recorded in the study database(s) or on electronic transcripts, nor on any other paper documents (apart from the informed consent form) or electronic files. Household addresses, geographic coordinates of households, participant name and other contact data for each household or participant will be kept separate and limited to authorized staff at the study site(s) only. Instead, household addresses or geographic coordinates, participant names and contact data will be replaced by a specific study household and subject identification code (=pseudonym). These pseudonymized data will be used during the study.

Thus we will be able to compile the prevalence rate at the time of the third and final screening round. In the analysis we will use specific survey statistics, taking into account the probability of being sampled for individuals in each of the study villages. Data management processes will be set up in compliance with ICH-GCP guidelines. With regard to personal data processing, we will adopt the principles of data minimization, proportionality of data collection and anonymize or pseudonymize personal data at the earliest convenience. We will also adopt Privacy by Design in the data management setup and perform a Data Protection Impact Assessment prior to any personal data collection, in compliance with the EU GDPR regulation.

3.5 Please give details on how reporting obligations (regarding study initiation, safety of study participants, ethical concerns, quality issues, integrity of data, study results) to regulatory bodies and ethics committees will be met.

There will be yearly reports to ethics committees in DRC in which study progress will be reported. Study protocol amendments (also regarding safety) will be submitted for approval to the respective EC's and RAs as well. Data integrity will be monitored by the Clinical Trials Unit of DNDi. A detailed data management plan as well as a statistical analysis plan will be drawn up at the start of the study, before the first participant is dosed. Safety of acoziborole will be monitored by DNDi. DNDi will report to the scientific advisory board on a 1 year - basis or ad hoc in case of any serious adverse event.

- 3.6 Please list all items of the sponsor's responsibilities (e.g. monitoring clinical sites, meeting regulatory obligations, data management, etc.) that will be supported by entities that are not part of the sponsor's organisation. Please describe how the sponsor will ensure oversight of these activities.

The sponsor of the clinical trial (phase 3b trial on safety of acoziborole) will be DNDI. DNDi has a dedicated unit for conducting and monitoring clinical trials. The design and implementation of the trial will be managed by DNDI in close collaboration with ITM and PNLTHA. This includes setting up and implementing the data collection system. The clinical trial units of ITM and DNDi will monitor data collection and data integrity and will also monitor trial implementation in general, in accordance with GCP standards.

- 3.7 What are the plans for major study milestones and what evidence supports its feasibility?

Please describe a realistic plan (based on prior experience) detailing the time necessary for (i) compiling the required regulatory and ethics submission package, (ii) receipt of regulatory and ethics approval, (iii) initiation of clinical site(s), (iv) completion of recruitment of the study population, (v) final assessment of all study participants, (vi) analysis and reporting of the study results.

During the first 6 months of the project (3rd and 4th quarter 2023) we will finalize the study protocol and obtain ethics and regulatory approval. The consortium partners, in particular ITM, IRD and DNDi, have ample experience in obtaining ethics approval for large field trials. In the meantime we will map the study area and organize refresher trainings in screening procedures for fixed health facilities to be involved in the study. There will be an official Initiation visit performed at the provincial gHAT coordination by the sponsor before the start of recruitment.

The villages will be screened for a first time in 2023/Q12024 and again in 2025 and in 2026, passive screening from fixed health facilities will be provided in parallel. The study population will be recruited among those testing positive to the serological screening test used. We foresee screening approximately 150,000 persons in 2024 and identify 2-3,000 serological suspects, i.e. 10-15 per working day. During the second and third year we expect to identify 1,000 additional serological suspects, excluding those already treated in earlier years. Active screening will be performed by three mobile teams, the standard for these mobile teams is to cover 60,000 population per year. Serological suspects recruited by the mobile team will be visited in their homes by a study physician who will assess them for eligibility, administer the study treatment, ensure their daily observation for safety for 5 days and ensure further follow-up.

In the 4th year, we will screen not just the endemic villages ($n = \pm 150,000$) but also a random sample from the other villages ($n = \pm 150,000$) and provide diagnostic confirmation on the spot for serological suspects. For this purpose three additional mobile teams will be involved.

Throughout the period of screening activities, samples will be collected and forwarded to the laboratory in Kinshasa. If we assume 3,000 samples per year, the number to be tested per working week would be in the order of 50-100. For ELISA or inhibition ELISA this is equivalent to one or two plates per week, which is not a major workload. We expect less than 10% to test positive to these immunological tests and thus require immune trypanolysis and molecular testing, i.e. less than 10 per week.

A study Close-Out visit will be performed by the sponsor once all study activities on site have been completed. Following this the database will be cleaned and locked for final analysis. We expect to have all results available by the end of the final screening year, 2027. This will leave sufficient time for the final analysis before the closure of the project foreseen for the end of the 2nd quarter of 2028.

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August 5, 2022

To whom it may concern:

I am writing in support of the Global Health EDCTP3 proposal “Stop transmission of gambiense human African trypanosomiasis (STROGHAT)”, led by ITM. This proposal will pilot the use of acoziborole in a “screen & treat” approach for the elimination of gambiense human African trypanosomiasis (HAT). The project will assess the feasibility, effectiveness, and cost of screen & treat with acoziborole for eliminating HAT transmission while also providing additional safety data on the use of acoziborole in seropositive individuals under the guidance of DNDi.

At the Bill & Melinda Gates Foundation we believe that a screen & treat strategy with acoziborole is an important tool for accelerating elimination of HAT transmission. Screen & treat overcomes the many of the challenges (including suboptimal sensitivity) associated with currently available confirmatory tests and increases the likelihood that cryptic human infections (such as dermal-only infections) are appropriately treated, reducing onward transmission. Mathematical models of HAT transmission suggest that screen & treat can be a cost-effective way to eliminate transmission; this project will be proof of principle.

Our foundation has invested heavily in all aspects of HAT elimination, and the proposed STROGHAT project will complement and support our activities. ITM and DNDi are trusted partners with deep and broad expertise in HAT. We have provided significant funding for the development of acoziborole, and we fully endorse this proposal to explore the optimal use of this game-changing drug.

Please feel free to contact me if you have any questions.

Kindest regards,



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